

3.1 General properties of waves — 5054 Paper 2

2015 onwards. Only 3.1 subquestions. Parts from the same question that are optics, EM-spectrum uses, hearing range, ultrasound applications, TIR or motors are omitted.

2015

May/June 5054/21

Question 4 • included (a), (b), (c)

- (a) State the unit in which pressure is measured.

.....[1]

- (b) Fig. 3.2 shows the axes for a graph of pressure against volume for the air in the syringe.

One point is plotted on the graph at pressure of P_0 and volume V_0 .

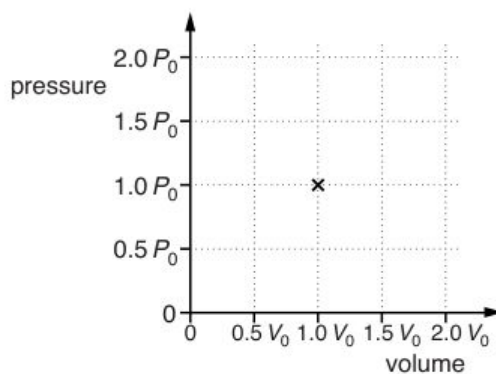


Fig. 3.2

The temperature of the air is kept constant.

On Fig. 3.2,

- (i) plot points at volumes of $0.5 V_0$ and $2.0 V_0$, [1]
(ii) complete the graph. [1]

(c) More metal weights are placed on top of the syringe.

Explain how the molecules of air inside the syringe are able to support more metal weights.

.....

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.....

.....

.....

.....

.....[3]

- 4 Fig. 4.1 shows a wave on a string. The wave is travelling towards the right.

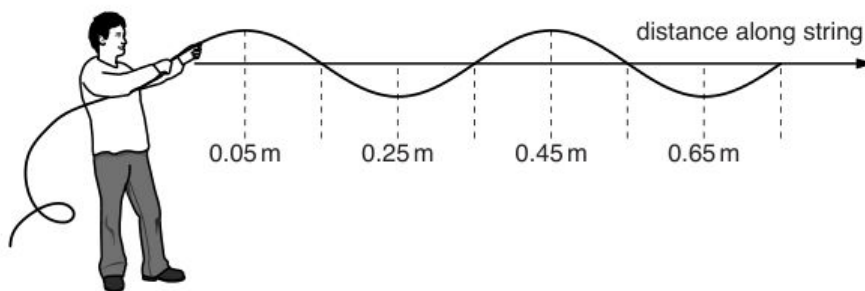


Fig. 4.1

- (a) Describe the movement of the particles of the string.

.....

[2]

- (b) Determine the wavelength of the wave.

wavelength =[1]

- (c) The speed of the wave is 2.0 m/s.

- (i) Calculate the frequency of the wave.

frequency =[2]

- (ii) On Fig. 4.1, draw the string at a time 0.10 s later than the time in Fig. 4.1. [1]

Mark scheme — 5054/21 Q4 (a), (b), (c)

4 PHYSICS 5054/21 Paper 2 (Theory), maximum raw mark 75 This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers. Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers. Cambridge will not enter into discussions about these mark schemes. Cambridge is publishing the mark schemes for the May/June 2015 series for most Cambridge IGCSE®,

Page 2	Mark Scheme	Syllabus	Paper
	Cambridge O Level – May/June 2015	5054	21
1	(a) (i) 60 m		B1
	(ii) 12 s		B1
	(b) (i) straight line from origin to 200 m at 40 s any line straight or curved from (40,200) to (60,500)		B1 B1
	(ii) $s = d/t$ or $500/60$ 8.3 m/s		C1 A1
2	(a) (i) force moves through a distance (in same direction)		B1
	(ii) chemical (potential) energy		B1
	(b) (i) 480 Nm		B1
	(ii) attempt to apply moments with two forces and distances 400 N		C1 A1
3	(a) Pa or N/m^2 or cm of mercury or atmosphere(s)		B1
	(b) correct points plotted at $(0.5V_0, 2P_0)$ and $(2V_0, 0.5P_0)$ curve through points of decreasing gradient		B1 B1
	(c) molecules hit sides/piston		B1
	more molecules hit per second/hit more frequently		B1
	molecular impacts create large(r) force (upwards on piston)		B1
4	(a) oscillate/vibrate stated or described transverse movement described		B1 B1
	(b) 0.40 m		B1
	(c) (i) $v = f\lambda$ or $(f=) v/\lambda$ or $2/(b)$ 5.0 Hz		C1 A1
	(ii) clear attempt to draw wave moved along 0.20 m to right		B1
5	(a) $\sin i/\sin r$ or $\sin 50/\sin 30$ 1.5(321)		C1 A1

(a) Explain how rubbing causes the balloon to become negatively charged.

.....
.....
..... [2]

(b) Explain why the hair is pulled towards the balloon.

.....
.....
.....
..... [2]

(c) Explain why it is important that the balloon is made from an electrical insulator.

.....
..... [1]

(d) State one example where static electricity is useful.

.....
..... [1]

- (a) The coil rotates and an alternating electromotive force (e.m.f.) is induced in the coil.

Fig. 6.2 shows how the alternating e.m.f. varies with time as the coil rotates.

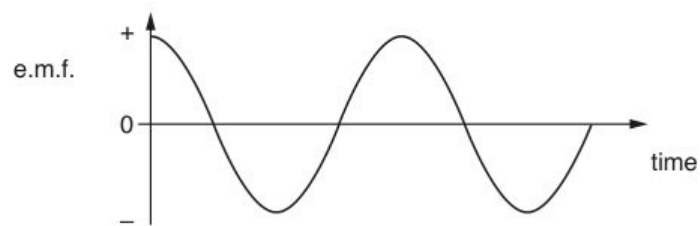


Fig. 6.2

Explain

- (i) why an e.m.f. is induced,

.....
.....
..... [2]

- (ii) why the e.m.f. is sometimes positive and sometimes negative.

.....
.....
..... [1]

(b) Changes are made to the a.c. generator, one at a time:

- stronger magnets are used
- more turns are wound on the coil
- the coil is turned faster.

Complete the table in Fig. 6.3 to show what happens to the maximum value of the e.m.f. and to the frequency of the alternating e.m.f.

changes made	what happens to the maximum value of the e.m.f.	what happens to the frequency of the e.m.f.
stronger magnets		
more turns on the coil		
the coil is turned faster		

Fig. 6.3

[3]

(a) State the purpose of the neutral wire.

.....
..... [1]

- (b) The live wire in the electric heater touches the outer metal case.

Explain how the earth and the fuse together protect the user from electric shock.

.....

.....

.....

..... [2]

- (c) The hairdryer does not have an earth wire. Explain why this hairdryer is still safe to use.

.....

..... [1]

- (d) In some modern homes, circuit breakers are used instead of fuses.

Suggest one advantage of using a circuit breaker rather than a fuse.

.....

..... [1]

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5054/22/M/J/15

- (a) Complete Fig. 8.1 to show the number of protons and neutrons in one nucleus of each of these isotopes of hydrogen.

	number of protons	number of neutrons
${}^1_1\text{H}$		
${}^2_1\text{H}$		

Fig. 8.1

[2]

- (b) Explain, using ideas about electrons, neutrons and protons, why atoms of ${}^1_1\text{H}$ and ${}^2_1\text{H}$ are uncharged.

.....

.....

.....

.....

.....

..... [3]

Mark scheme — 5054/22 Q9 (a), (b), (c), (d)

95 000/120 000 ($\times 100$) 0.79(17) or 79(.17)% A1 (ii) (P=) energy/time or 90 000 / 60 C1 1500 W A1 (b) electric kettle and B1 • more energy / heat per minute output / into water / supplied • more power output transfers heat / energy faster / at a faster rate (c) steam molecules have more potential energy; further apart; smaller force / bonds between molecules; have latent heat; more random arrangement B1 4 (a) (H=) mCT or $330 \times 4.2 \times 13$ C1 18 000 J or 18 020 J or 18 018 J A1 Page 3 Mark Scheme Syllabus Paper Cambridge O Level – May/June 2015 5054 22 © Cambridge International Examinations 2015 (b) ice takes in / needs heat / energy B1 • for latent heat • to melt / turn to water (at 0 °C) / change state • to break bonds / for

Page 4	Mark Scheme	Syllabus	Paper
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- 7 (a) to provide a complete circuit (with live) B1
or to pass current back to mains
or provide a return path for the current
- (b) current / charge / electrons flow to earth / earth wire / ground (when live touches case) B1
fuse melts / blows **and** disconnects circuit / cuts live / stops current B1
- (c) doubly insulated B1
or case/body made of plastic / insulator / not made of metal
or user cannot touch metal
- (d) (circuit breaker) B1
 - turns off / acts fast(er)
 - can be reset
 - easy to see it has tripped / switched
 - can detect small difference between live and neutral currents / small (leakage) current to earth
- 8 (a) left column both 1 B1
right column 0 **and** 1 B1
- (b) (at least one of the atoms) contain same number of electrons and protons B1
or have 1 electron and 1 proton
charge on electron and proton opposite B1
or electron negative **and** proton positive
or charge on electron neutralises/cancels/balances proton charge
neutrons have no charge B1
- 9 (a) number of waves (that pass a point) M1
or number of oscillations (passing a point)
in unit time **or** per second **or** in 1 second A1
- (b) (i) 1.5 cm B1
- (ii) $(v =)f\lambda$ **or** 5×1.5 seen C1
or $(s =)d/t$ **and** $f = 1/t$
7.5 cm/s A1
- (c) (i) wavelength decreases B1
travels a shorter distance in the same time B1
or frequency stays the same (and $v = f\lambda$)

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Oct/Nov 5054/21

Question 7 • included (a), (b)

not included: (c) use of ultrasound

(a) Define *boiling point*.

.....
..... [1]

(b) (i) Describe the molecular structure of a liquid.

.....
.....
.....
.....
..... [2]

(ii) Explain, using ideas about molecules, why thermal energy (heat) must be supplied to the water as it boils.

.....
.....
.....
.....
..... [2]

(a) Define the term *focal length*.

.....
.....
..... [1]

- (b) (i) On Fig. 6.1, mark and label with an F, each of the two focal points (principal foci) of the lens. [1]

- (ii) The point Y is the tip of the image.

On Fig. 6.1, draw a ray diagram to locate the position of the top of the object.
Label this point X. [3]

- (iii) Using Fig. 6.1, determine the distance of the candle from the centre of the lens.

distance = [1]

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- 7 Audible sound waves and ultrasound waves both consist of compressions and rarefactions.

- (a) Describe one difference between a compression and a rarefaction.

.....
.....
..... [1]

- (b) In a particular medium, the speed of both audible sound and ultrasound waves is v .

- (i) State the equation that relates the wavelength λ of a sound wave to its frequency f .

.....
..... [1]

- (ii) Explain how, in this medium, the wavelength of an audible sound wave compares with the wavelength of an ultrasound wave.

.....
.....
.....
..... [1]

Mark scheme — 5054/21 Q7 (a), (b)

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Page 3	Mark Scheme	Syllabus	Paper
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5	(a) temperature at which a liquid becomes a gas	B1	
	(b) (i) molecules close together / touching or closer than in gas randomly arranged or irregular structure	B1 B1	
	(ii) to separate / increase the distance between molecules work done against (intermolecular) forces or supply p.e. or break bonds	B1 B1	[5]
6	(a) distance from (optical) centre to focal point (principal focus)	B1	
	(b) (i) both Fs correctly positioned at ± 1 mm	B1	
	(ii) two of: paraxial ray to lens through focal point to image ray through optical centre ray through focal point and then paraxial to image (ign. arrows) X at crossing point of rays	M2 A1	
	(iii) 3.4–3.8 cm	B1	[6]
7	(a) at compression: molecules closer together or pressure higher or vice versa for rarefaction	B1	
	(b) (i) $v = f\lambda$ or in words	B1	
	(ii) larger and because the frequency is lower	B1	
	(c) states one use basic idea	(e.g. prenatal scanning) (e.g. ultrasound reflects off foetus)	B1 B1 [5]
8	(a) (i) $(I =)V/R$ or $12/(6000 + 2000)$ or $12/8000$ or $12/2000$ or $12/6000$ or in (ii) $(V =)IR$ or 0.0015×6000 1.5 mA	C1 A1	
	(ii) 9.0 V	B1	
	(b) (reading/it) increases resistance of LDR falls	B1 B1	
	(c) light meter / sensor or automatic light switch or something sensible	B1	[6]
			[45]

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May/June 5054/21

Question 5 • included (a), (b)

- (a) The student plots a graph of the extension of the spring against the mass hanging on the spring. Fig. 2.2 shows the student's graph.

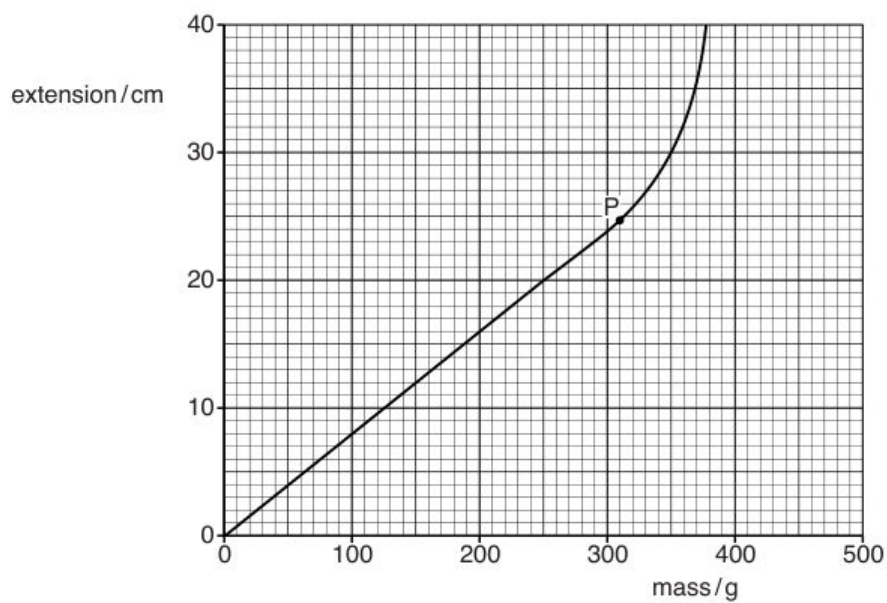


Fig. 2.2

(b) An identical spring is used with the original spring, as shown in Fig. 2.3.

Together they support the mass X.

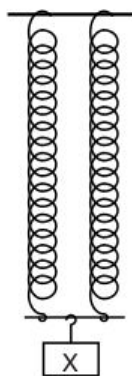


Fig. 2.3 (not to scale)

State and explain how the extension in Fig. 2.3 compares with the extension in Fig. 2.1.

.....

.....

.....

.....[2]

- (a) (i) State what is meant by *mass*.

.....
.....[1]

- (ii) The student is told that the density of the plastic is 0.91 g/cm^3 .

Calculate the volume of the plastic sheet.

volume =[2]

- (iii) The student measures the length and width of the sheet. The readings obtained are:

length of sheet = 3.0 cm

width of sheet = 2.0 cm

Calculate the thickness of the sheet.

thickness = [2]

- (b) State a measuring instrument that can be used to measure the thickness of the sheet accurately.

.....[1]

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5054/21/M/J/16

- 4 (a) Equal volumes of air, copper and water are heated from 20°C to 80°C . They expand by different amounts.

Draw a line from each box on the left-hand side to show which expands the most and which expands the least.

expands the most

air

copper

expands the least

water

[1]

(b) Fig. 4.1 shows a mercury thermometer.

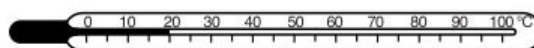


Fig. 4.1

This thermometer has a suitable range for use in laboratory experiments, but has a low sensitivity for some experiments.

(i) With reference to this thermometer, describe what is meant by

1. *range*,

.....
.....
.....[1]

2. *low sensitivity*.

.....
.....
.....[1]

(ii) State one change in the design of a thermometer that increases its sensitivity.

.....
.....[1]

Mark scheme — 5054/21 Q5 (a), (b)

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Page 3	Mark Scheme	Syllabus	Paper
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- (b) (i) 1 difference between smallest and largest temperature
or from 0 to 100 °C B1
- (i) 2 small/moderate distance between (thermometer) marks B1
or for a given temperature change there is a small expansion of liquid / distance
(along scale) / change in thermometric property
or cannot measure small temperature difference / change
- (ii) • use liquid that expands more B1
• smaller bore / thinner tube
• more mercury (in bulb) or use larger bulb
- 5 (a) *sound*: along or parallel (to transfer of energy or wave) **and** longitudinal B1
water: perpendicular **and** transverse B1
- (b) (i) 0.29 – 0.28 m B1
- (ii) time / period for one wave(length) / cycle constant B1
or each oscillation / cycle takes one second
- 6 (a) angle of incidence B1
smallest angle for light to be totally internally reflected B1
or largest angle (of incidence) for ray to be refracted / emerge
or when light emerges along surface
or when angle of refraction is 90°
- (b) (i) $n = 1 / \sin C$ algebraic or numerical C1
2.5 or 2.46 or 2.458(59) A1
- (ii) *left hand diagram* ray refracts away from normal and emerges into air at bottom left surface B1
right hand diagram reflected horizontal ray (by eye) B1
right hand diagram rest of ray completely correct to emerge into air at top face without refraction (by eye) B1
- 7 (a) (current in coil) creates magnetic field C1
or current is at right angles to magnetic field (of permanent / cylindrical magnets)
- (b) into and out of magnet B1
or left and right
or backwards and forwards
current is one way then reverses (so reverses force) B1

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Question 7 • included (c)

only (c) $v = f\lambda$ • not (a)(b) motor

- (c) The loudspeaker produces sound of frequency 500Hz. The speed of sound is 320m/s. Calculate the wavelength of the sound produced.

wavelength =[2]

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5054/21/M/J/16

Mark scheme — 5054/21 Q7 (c)

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or from 0 to 100 °C B1
- (i) 2 small/moderate distance between (thermometer) marks B1
or for a given temperature change there is a small expansion of liquid / distance
(along scale) / change in thermometric property
or cannot measure small temperature difference / change
- (ii) • use liquid that expands more B1
• smaller bore / thinner tube
• more mercury (in bulb) or use larger bulb
- 5 (a) *sound*: along or parallel (to transfer of energy or wave) **and** longitudinal B1
water: perpendicular **and** transverse B1
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- (ii) time / period for one wave(length) / cycle constant B1
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- 6 (a) angle of incidence B1
smallest angle for light to be totally internally reflected B1
or largest angle (of incidence) for ray to be refracted / emerge
or when light emerges along surface
or when angle of refraction is 90°
- (b) (i) $n = 1 / \sin C$ algebraic or numerical C1
2.5 or 2.46 or 2.458(59) A1
- (ii) *left hand diagram* ray refracts away from normal and emerges into air at bottom left surface B1
right hand diagram reflected horizontal ray (by eye) B1
right hand diagram rest of ray completely correct to emerge into air at top face without refraction (by eye) B1
- 7 (a) (current in coil) creates magnetic field C1
or current is at right angles to magnetic field (of permanent / cylindrical magnets)
- (b) into and out of magnet B1
or left and right
or backwards and forwards
current is one way then reverses (so reverses force) B1

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Question 5 • included (a), (b)

not included: (c) sound production / vacuum

- (a) Explain what is meant by the term *critical angle*.

.....

.....

.....

.....[2]

- (b) The lamp sends light towards the surface of the pool.

Fig. 4.1 shows three rays of light that are at 30° , 60° and 90° to the horizontal.

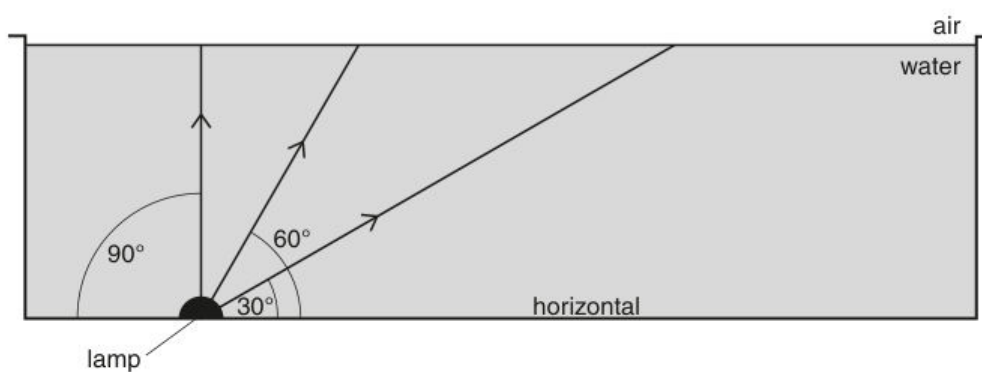


Fig. 4.1

On Fig. 4.1, draw the path taken by each of the three rays after they strike the surface of the water. [3]

- 5 Fig. 5.1 shows a loudspeaker inside a glass bell-jar that contains air.

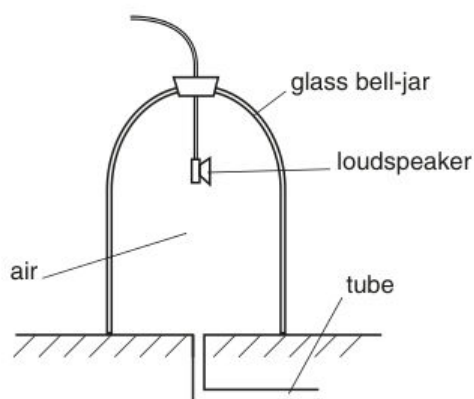


Fig. 5.1

The loudspeaker produces a sound of frequency 2200 Hz.

- (a) State what is meant by the *frequency* of a sound.

.....
.....
.....[1]

- (b) (i) The speed of sound in air is 330 m/s.

Calculate the wavelength of the sound in the air.

wavelength =[2]

- (ii) The sound passes from the air into the glass.

State what happens to

1. the frequency of the sound,

.....

2. the speed of the sound.

.....
[1]

Mark scheme — 5054/21 Q5 (a), (b)

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5	(a) number of oscillations / vibrations / wavelengths / compressions / rarefactions / cycles per second / unit time	B1	
	(b) (i) $(\lambda =)c/f$ or 330 / 2200 0.15 m	C1 A1	
	(ii) 1 no change and 2 increases	B1	
	(c) (i) 1 loudspeaker vibrates / oscillates / moves to and fro (and collides with molecules) 2 compressions and rarefactions / molecules vibrate / longitudinal wave vibration / oscillation / energy passed on	B1 B1 B1	
	(ii) fewer / no molecules / particles and less / no energy / vibration transferred	B1	[8]
6	(a) (i) X N-pole Y S-pole and Z N-pole	B1 B1	
	(ii) they touch / move towards each other and opposite poles attract	B1	
	(b) any sensible use: starting-motor circuit; with a logic gate; nuclear power station corresponding explanation: current too large for dash-board switch; current too small to power device; too dangerous to reach switch	B1 B1	[5]
7	(a) (i) supplies the (mains) e.m.f. / voltage	B1	
	(ii) to complete the circuit / is at 0 V	B1	
	(b) (i) the circuit / supply is cut / broken or current stops fuse melts / blows / burns	B1 B1	
	(ii) live wire when it cuts the circuit / melts no part of the appliance is live / no shock	B1 B1	[6]
[45]			

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Oct/Nov 5054/22

Question 10 • included (a)

- (a) Calculate the speed of the skier after 5.0 s.

speed =[2]

- (a) The skateboarder moves off and she descends vertically through 1.8 m. The gravitational field strength g is 10 N/kg.

- (i) Calculate the change in gravitational potential energy of the skateboarder.

energy change =[2]

- (ii) State the name of two forms of energy that increase as she descends.

1.

2.

[2]

- (a) The gravitational field strength g is 10 N/kg.

- (i) Calculate the weight of the lid.

weight =[1]

- (ii) The lid is uniform and its centre of mass is at its centre. The weight of the lid produces a moment about the hinge.

1. Calculate the moment about the hinge when the lid is closed.

moment =[2]

2. The moment required to open the lid is greater than the value calculated in (a)(ii)1.

Suggest one reason for this.

.....

.....[1]

- (a) State what is meant by *boiling point*.

.....

.....[1]

Mark scheme — 5054/22 Q10 (a)

10×1.8 B1 810 J B1 (ii) kinetic either order B1 thermal / internal / heat / sound B1 (b) (i) upwards / centripetal / towards centre (of circle) B1 (ii) it / weight less (than normal contact force) or upward force greater B1 [6] 3 (a) (i) 20 N B1 (ii) 1. ($\Gamma =$) Fd or 20×0.35 or 20×0.70 or 14 C1 7.0 N m A1 2. friction (at hinge / seal) or air resistance or to cause an initial acceleration B1 (b) (for other directions) perpendicular distance is less B1 [5] 4 (a) temperature at which liquid / water turns to gas / vapour / steam B1 (b) (i) ($T =$) 24 ($^{\circ}\text{C}$) or 100 – 24 or 76 C1 ($\Delta Q =$) $mc\Delta T$ or $1.5 \times 4200 \times 76$ C1 4.8×105 J A1 (ii) heat is lost (to the surr

Page 5	Mark Scheme		Syllabus	Paper
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	(ii) 1	cylinder is magnetised (by induction) top (of cylinder) is an S-pole unlike poles attract or S-pole attracts N-pole	B1 B1 B1	
	2	it does not (remain in contact) and iron is temporary / soft magnetic material / core (and cylinder) lose magnetisation	B1	[4]
				[15]
10 (a)	(i)	$3.0 \times 10^8 \text{ m/s}$	B1	
	(ii)	$(\lambda =)c / f$ or $3.0 \times 10^8 / 4.3 \times 10^{14}$ $7.0 \times 10^{-7} \text{ m}$	C1 A1	[3]
	(b) (i)	decreases	B1	
	(ii)	$\sin(i) = n \times \sin(r)$ or $1.5 \times \sin(30^\circ)$ or 0.75 49°	C1 A1	
	(iii)	41°	B1	[4]
	(c) (i)	dispersion at both surfaces and refractions in correct direction violet / blue light below the red light shown	B1 B1	
	(ii)	spectrum or band of (continuous) colours or colours of rainbow red, orange, yellow, green, blue, (indigo, violet)	B1 B1	
	(iii) 1	X marked above red	B1	
	2	it is / black surfaces are good absorbers (of IR radiation)	B1	[6]
	(d)	intruder / human being emits IR	B1	
		or		
		intruder warm or IR detected	B1	
		or		
		does not reach detector		
		or		
		change detected	B1	[2]
				[15]

2017

May/June 5054/21

Question 10 · included (b), (c)

not included: (a) plane mirror; (c)(ii) hearing range

- (b) Complete the equation to show the radioactive decay of $^{131}_{53}\text{I}$.



[2]

- (c) When a nucleus of $^{125}_{53}\text{I}$ decays, only a gamma ray is emitted.

Suggest why it is better for the doctor to inject the isotope $^{125}_{53}\text{I}$ into the patient.

.....

.....

.....

.....[2]

(b) Fig. 9.1 is a block diagram of a power station that produces electrical energy from oil.

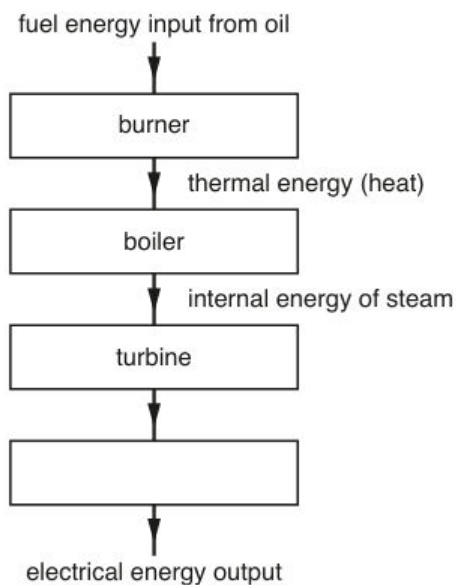


Fig. 9.1

(i) Write the name of the missing part of the power station in the empty box on Fig. 9.1. [1]

(ii) State the form of energy that the turbine possesses.

.....[1]

(iii) A small boiler in the power station contains 24 m^3 of water at 30°C . High pressure in the boiler increases the boiling point of water to 120°C .

Thermal energy supplied to the boiler is used to heat the water from 30°C to 120°C and then to turn it all to steam at 120°C .

The density of water is 1000 kg/m^3 .

The specific heat capacity of water is $4200\text{ J/(kg}^\circ\text{C)}$.

The specific latent heat of vaporisation of water is $2.3 \times 10^6\text{ J/kg}$.

1. Calculate the mass of water in the boiler.

mass =[1]

2. Calculate the total thermal energy (heat) supplied to the boiler.

thermal energy =[4]

- (iv) The electrical energy output from the power station is transmitted over long distances at a high voltage.

Explain why electricity is transmitted at a high voltage.

.....
.....
.....
.....[2]

(b) Fig. 10.2 shows wavefronts in a ripple tank. They move in the direction of the arrow.

The wave hits the boundary between two regions and the wave slows down as it enters the shaded region.

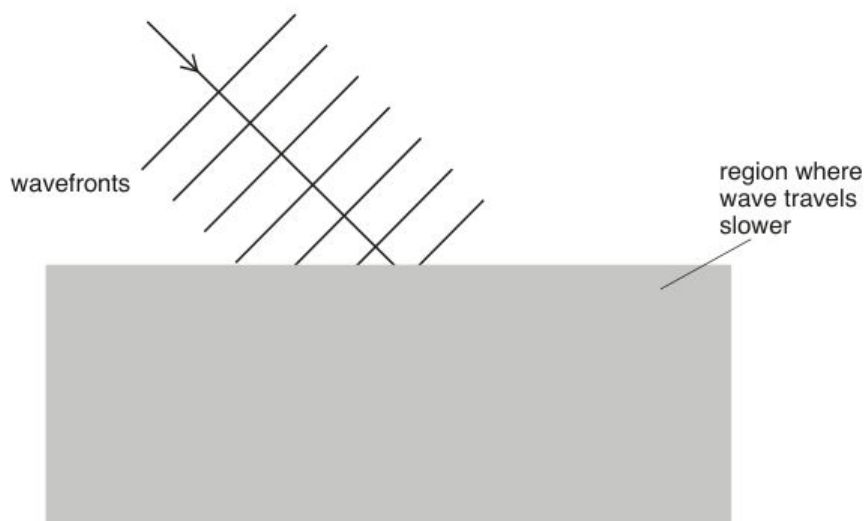


Fig. 10.2

(i) State what is meant by *wavefront*.

.....

.....[2]

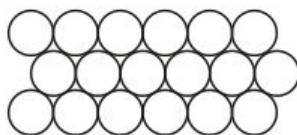
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5054/21/M/J/17

Mark scheme — 5054/21 Q10 (b), (c)

10(b)(i) line (joining points) C1 line joining points / particles on wave with same phase or line joining points along a crest etc. A1 10(b)(ii)1 correct angle to surface $\pm 7^\circ$ with correct orientation and similar wavelength (by eye) B1 10(b)(ii)2 at least two lines at smaller angle to surface with correct orientation in slower medium B1 showing smaller and constant wavelength with wavefronts deviated in correct direction B1 10(c)(i) $(v =) f\lambda$ or $2(000) \times 16$ C1 32 000 cm / s or 320 m / s A1 10(c)(ii)1 a range with 15–25 Hz as the lowest frequency and 15–30 kHz as the highest B1 10(c)(ii)2 1.6 cm (if highest frequency is 20 000 Hz) B1 10(c)(iii) tube or other method to produce (narrow) beam of sound on source and/or on detector B1 stated detector or stated reflector B1 detector moved to find maximum loudness or angles i and r measured with suitable experiment B1 5054/21 – Mark Scheme PUBLISHED May/June 2017 Page 8 of 8 Question Answer Marks

Question	Answer	Marks
10(a)(i)	both reflected rays correct by eye	B1
	image in correct position shown by continuation of rays behind mirror	B1
10(a)(ii)	virtual or upright or laterally inverted or same size (as object)	B1
10(b)(i)	line (joining points)	C1
	line joining points / particles on wave with same phase or line joining points along a crest etc.	A1
10(b)(ii)1	correct angle to surface $\pm 7^\circ$ with correct orientation and similar wavelength (by eye)	B1
10(b)(ii)2	at least two lines at smaller angle to surface with correct orientation in slower medium	B1
	showing smaller and constant wavelength with wavefronts deviated in correct direction	B1
10(c)(i)	$(v =) f\lambda$ or $2(000) \times 16$	C1
	32 000 cm / s or 320 m / s	A1
10(c)(ii)1	a range with 15–25 Hz as the lowest frequency and 15–30 kHz as the highest	B1
10(c)(ii)2	1.6 cm (if highest frequency is 20 000 Hz)	B1
10(c)(iii)	tube or other method to produce (narrow) beam of sound on source and/or on detector	B1
	stated detector or stated reflector	B1
	detector moved to find maximum loudness or angles i and r measured with suitable experiment	B1

May/June 5054/22**Question 6 · included (a), (b)***not included: (b)(ii) hearing range***(a)** Fig. 5.1 shows the arrangement of the molecules in the solid.**Fig. 5.1**

In the space below, draw the arrangement of the molecules in the liquid.

[1]

(b) Complete the table by describing the motion of the molecules in the solid, liquid and gas.

	motion of the molecules
solid	
liquid	
gas	

[3]

6 (a) Explain the difference between a longitudinal wave and a transverse wave.

You may draw a diagram, if you wish, to help your explanation.

.....

.....

.....

.....

.....[2]

(b) A sound wave has a frequency of 3.8 kHz and a speed of 330 m/s.

(i) Calculate the wavelength of the sound.

wavelength =[2]

(ii) A different sound has a frequency of 3.8 Hz.

State and explain whether a human with normal hearing is able to hear this sound.

.....
.....
.....[1]

Mark scheme — 5054/22 Q6 (a), (b)

6(a) longitudinal - vibration / oscillation / movement to and fro and in direction of wave or has compressions and rarefactions B1 transverse – vibration / oscillation / movement up and down and at right angles to wave or has crests and troughs B1 6(b)(i) ($\lambda =$) v / f or $330 / 3800$ B1 0.087 m or 8.7 cm B1 6(b)(ii) not heard and because below the range of audible frequencies or audible range is 20 – 20 000 Hz or too low a pitch / frequency B1 Question Answer Marks

Question	Answer	Marks
6(a)	longitudinal - vibration / oscillation / movement to and fro and in direction of wave or has compressions and rarefactions	B1
	transverse – vibration / oscillation / movement up and down and at right angles to wave or has crests and troughs	B1
6(b)(i)	$(\lambda =) v / f$ or 330 / 3800	B1
	0.087 m or 8.7 cm	B1
6(b)(ii)	not heard and because below the range of audible frequencies or audible range is 20 – 20 000 Hz or too low a pitch / frequency	B1

Question	Answer	Marks
7(a)(i)	current in coil (at right angles) in a magnetic field (of magnet) or left-hand rule mentioned	B1
7(a)(ii)	reverses / changes direction of current (in coil)	B1
	reverses current every half turn / when coil is vertical or reverses forces (on side AB / CD) or keeps forces in same direction for wire on one side	B1
7(b)(i)	$(E =) VIt$ or $2 \times 12 \times 8$ or $E = Pt$ and $P = VI$ or $E = VQ$ and $Q = It$	C1
	190 or 192 J	A1
7(b)(ii)	73% or 0.73	B1

2018**May/June 5054/21****Question 11 · included (a), (b)****10 (a) (i)** Describe two differences between boiling and evaporation.

1.

.....

.....

2.

.....

.....

[4]

(ii) Explain, using ideas about molecules, why thermal energy is needed to boil a liquid.

.....

.....

.....

.....[2]

- (b) In one type of bathroom shower, cold water passes through a metal pipe which contains an electric heater.

The cold water is heated and emerges from the shower head.

The temperature of the cold water before heating is measured and the hot water emerging from the shower in 1.0 minute is collected in a container.

Measurements and other data are:

temperature of water before heating = 16°C

temperature of water after heating = 37°C

volume of water collected in 1.0 minute = $4.6 \times 10^{-3} \text{ m}^3$

specific heat capacity of water = $4200 \text{ J/(kg }^{\circ}\text{C)}$

density of water = 1000 kg/m^3

- (i) Calculate the mass of water leaving the shower in 1.0s.

mass =[2]

- (ii) Calculate the thermal energy (heat) gained by the water in 1.0 s.

energy =[3]

- (iii) To calculate the efficiency of the shower, the value in **(b)(ii)** and the value of one other quantity are needed.

1. State what other quantity is needed.

.....

2. Describe how these two values are used to calculate the efficiency of the shower.

.....

[2]

- (iv) The metal pipe that contains the electric heater is earthed.

Explain why this is necessary.

.....

[2]

11 A water wave is produced in a ripple tank.

- (a) (i) Draw a diagram of a ripple tank and describe how it is used to produce and observe the wave.

.....

.....

.....

.....

.....

.....

.....[4]

- (ii) The water wave in the ripple tank transfers energy.

Describe how to show that there is a transfer of energy.

.....

.....

.....

.....[2]

- (iii) State what is meant by a *wavefront*.

.....

.....

.....[2]

- (b) Fig. 11.1 shows the variation with time t of the height of water at the same place in the tank for two waves P and Q. The height is sometimes above and sometimes below the mean level which is marked as 0.

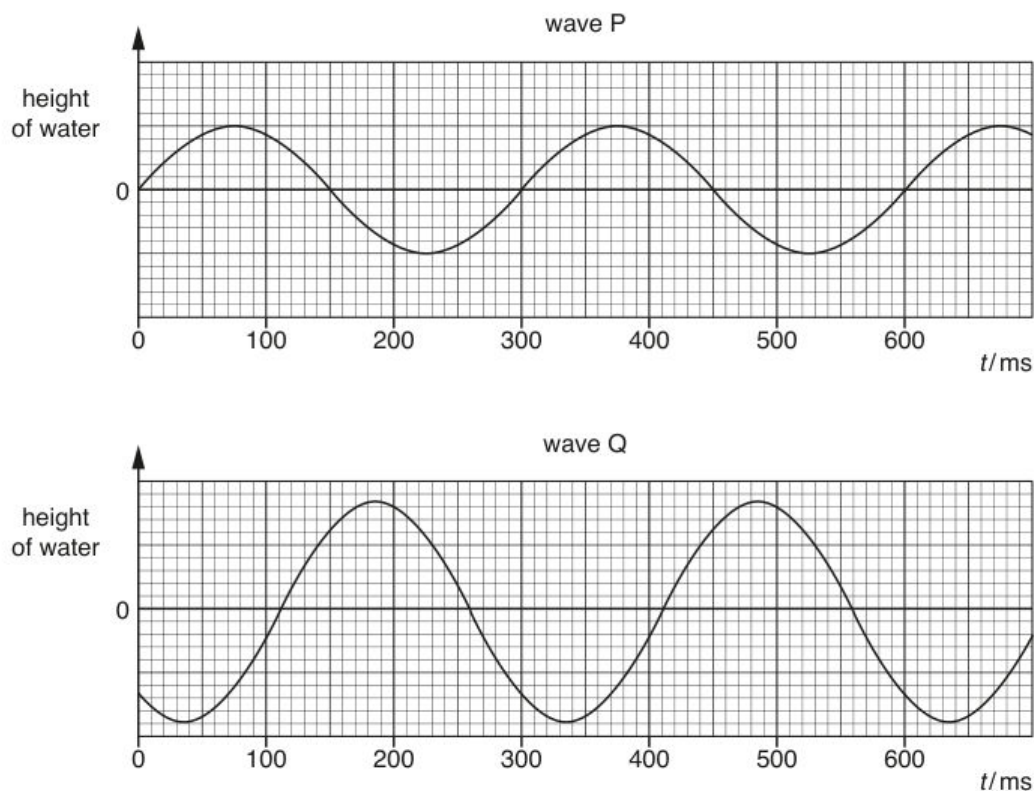


Fig. 11.1

- (i) State one difference and one similarity of the two waves.

difference

.....

similarity

.....

[2]

Mark scheme — 5054/21 Q11 (a), (b)

11(a)(i) tank with water B1 dipper (bar or point) touches or in surface of water B1 light source / stroboscope / video B1 dipper moves up and down (to create wave) B1 11(a)(ii) observe (surface of) water OR place particle / cork on surface OR equivalent movable / rotating object B1 moves up and down B1 11(a)(iii) line joining points M1 that are at the same point in the motion A1 11(b)(i) difference: Q has a greater / different amplitude or phase B1 similarity: same frequency; same time for one wave (period); both transverse; oscillation up and down B1 11(b)(ii) time 300 (ms) or 0.3 (s) C1 ($f = 1/t$) numerical or algebraic in any form C1 3.3 Hz A1 11(b)(ii) 2 ($\lambda = v/f$ or $v = \lambda f$) numerical or algebraic in any form C1 0.060 m A1

Question	Answer	Marks
11(a)(i)	tank with water	B1
	dipper (bar or point) touches or in surface of water	B1
	light source / stroboscope / video	B1
	dipper moves up and down (to create wave)	B1
11(a)(ii)	observe (surface of) water OR place particle / cork on surface OR equivalent movable / rotating object	B1
	moves up and down	B1
11(a)(iii)	line joining points	M1
	that are at the same point in the motion	A1
11(b)(i)	<i>difference</i> : Q has a greater / different amplitude or phase	B1
	<i>similarity</i> : same frequency; same time for one wave (period); both transverse; oscillation up and down	B1
11(b)(ii)	time 300 (ms) or 0.3 (s)	C1
	(f=) 1 / t numerical or algebraic in any form	C1
	3.3 Hz	A1
11(b)(ii)	2 ($\lambda =$) v / f or $v = \lambda f$ – numerical or algebraic in any form	C1
	0.060 m	A1

May/June 5054/22**Question 5 • included (b)***only (b) • not (a) satellite distance*

(b) The object has a constant speed for some of the time.

(i) State and explain how a distance-time graph shows that speed is constant.

.....
.....
.....
.....[2]

(ii) Explain, in terms of the forces acting on the object, how it is able to fall at a constant speed.

.....
.....
.....
.....[2]

- (b) Calculate the load that produces an extension of 49 mm.

load =[2]

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- (b) Fig. 3.1 is a diagram of the metal at 20 °C. The metal is a large, square sheet with a square hole at its centre. The sheet lies flat on a table.

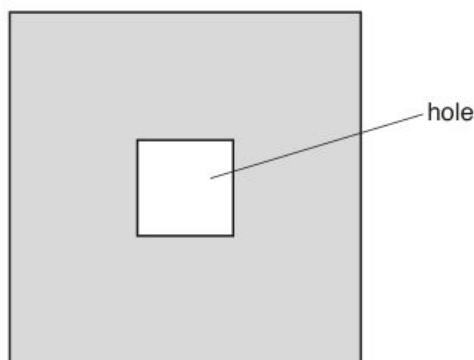


Fig. 3.1

On Fig. 3.1, draw the outline of the sheet and of the hole after the metal is heated. Make sure that the expansion is noticeable. [2]

Mark scheme — 5054/22 Q5 (b)

5(b)(i) 1st two columns correct (sound longitudinal and microwaves transverse) B1 3rd column correct microwaves electromagnetic B1 5(b)(ii) Layers / molecules / particles / spring coils close together or high(er) pressure (than atmospheric) B1 Question Answer Marks

Question	Answer	Marks
5(a)	(d=) s × t in any form algebraic or numerical,	C1
	3.6×10^7 m	A1
5(b)(i)	1st two columns correct (sound longitudinal and microwaves transverse)	B1
	3rd column correct microwaves electromagnetic	B1
5(b)(ii)	Layers / molecules / particles / spring coils close together or high(er) pressure (than atmospheric)	B1

Question	Answer	Marks
6(a)	any two of - ray <u>through</u> middle of lens undeviated - ray parallel to axis passes <u>through</u> focus, 3 cm from lens - ray <u>through</u> focus on left of lens parallel to axis after lens	B2
6(b)	inverted or real	B1
6(c)	image size / object size or image distance / object distance numerical or algebraic	C1
	rays from lens intersect on diagram (for image) and 1.3–1.7	A1
6(d)	projector or photographic enlarger	B1

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Question 9 • included (a), (b)

not included: (c) TIR / critical angle

- (a) On Fig. 3.2, sketch the shape of the graph for loads greater than 7.0 N. [1]

- (b) All the loads are removed from the spring. A block of wood is suspended from the end of the spring. The length of the spring increases from 4.9 cm to 13.4 cm.

Determine the weight of the wooden block.

weight = [2]

- (a) Complete the equation for the kinetic energy of an object travelling with a speed v and state what any other terms in the equation represent.

kinetic energy =

[1]

- (b) The mass of the rock is 0.60 kg. When it hits the Earth, its average temperature increases by 25°C .

- (i) The specific heat capacity of the rock is $560\text{ J}/(\text{kg}^{\circ}\text{C})$.

Calculate the increase in the internal energy of the rock.

increase in internal energy = [2]

- (ii) Use the answer from (b)(i) to determine the minimum possible speed of the rock when it hits the Earth.

minimum speed = [2]

- (iii) Suggest one reason why the speed determined in (b)(ii) is less than the actual speed of the rock when it hits the Earth.

.....
..... [1]

(a) The clinical thermometer has a small range and a high sensitivity.

(i) State what is meant by

1. *range*,

.....
..... [1]

2. *sensitivity*.

.....
..... [1]

(ii) State and explain one feature of this clinical thermometer that produces a high sensitivity.

.....
.....
.....
..... [2]

(b) State the purpose of the constriction in the clinical thermometer.

.....
..... [1]

(a) State the material from which the compass needle is made.

..... [1]

(b) Describe how the small compass is used to plot magnetic field lines in the region surrounding the magnet.

.....
.....
.....
.....
..... [3]

Mark scheme — 5054/21 Q9 (a), (b)

9(a) the vibration direction is parallel to the wave / energy travel direction or longitudinal waves have compressions and rarefactions or transverse waves have crests and troughs or longitudinal waves cannot be polarised B1 9(b)(i) ($f = v / \lambda$ or $0.17 / 0.019$ C1 8.9 Hz A1 9(b)(ii) (frequency) stays constant or does not change B1 9(b)(iii) (speed) decreases B1 (Fig. 9.1 shows that) wavelength decreases or refraction is towards normal or $i > r$ or top of wave (in shaded region) lags behind bottom of wave B1

Question	Answer	Marks
8(d)(ii)	from chemical (energy) or chemical (energy) as first term	B1
	to gravitational potential (energy) or to thermal / heat (energy) as last term	B1
	to thermal / internal (energy) / heat and thermal / internal (energy) / heat	B1
8(e)	no resultant force or forces balance / cancel or tension equals weight (of bucket) or upwards force equals downward force	B1

Question	Answer	Marks
9(a)	the <u>vibration</u> direction is parallel to the wave / energy travel direction or longitudinal waves have compressions and rarefactions or transverse waves have crests and troughs or longitudinal waves cannot be polarised	B1
9(b)(i)	$(f =) v / \lambda$ or 0.17 / 0.019	C1
	8.9 Hz	A1
9(b)(ii)	(frequency) stays constant or does not change	B1
9(b)(iii)	(speed) decreases	B1
	(Fig. 9.1 shows that) wavelength decreases or refraction is towards normal or $i > r$ or top of wave (in shaded region) lags behind bottom of wave	B1
9(c)(i)	electromagnetic and transverse underlined	B1
9(c)(ii)	$n = \sin(i) / \sin(r)$ or $(r =) \sin^{-1}(\sin(i) / n)$ or $(r =) \sin^{-1}(\sin(60^\circ) / 1.6)$	C1
	33(°)	A1

Question	Answer	Marks
9(c)(iii)	1 $n = 1 / \sin(c)$ or $(c =) \sin^{-1}(1 / n)$ or $\sin^{-1}(1 / 1.6)$	C1
	39°	A1
	2 total internal reflection	B1
	$\theta > c$ or 57(°) > 39(°) or angle of incidence greater than the critical angle	B1
	3 reflected ray at P (correct angle by eye)	B1
	light in air at 30° (by eye) to vertical surface	B1

Question	Answer	Marks
10(a)(i)	same number of protons or same number of electrons	B1
10(a)(ii)	different number of neutrons	B1
10(b)(i)	${}^4_2\alpha$	B1
	${}^{238}_{92}\text{U}$	B1
	${}^{235}_{92}\text{U}$	B1
10(b)(ii)	(no. of half-lives =) $1.2 \times 10^5 / 2.4 \times 10^4$ or 5 (half-lives)	C1
	relevant halving or $1 / 2^5$ or 1 / 32 or 6400 / 32	C1
	200 counts / second	A1

Question 9 · included (b), (c)

not included: (a) sound vs ultrasound

(c) The half-life of cobalt-60 is 5.3 years.

(i) State what is meant by *half-life*.

.....

.....

..... [2]

(ii) When there is a lead sheet between the detector and the sample, the average count rate is obtained from six readings taken at one-minute intervals. The six readings are given in the table in Fig. 6.4.

reading number	1	2	3	4	5	6
count rate counts/minute	61	46	43	56	49	57

Fig. 6.4

There are reasons for suggesting that the variation in these readings is random and not because the number of cobalt-60 atoms in the sample is decreasing.

State two of these reasons.

1.

.....

2.

..... [2]

(b) When there is an alternating current (a.c.) in the primary coil, the lamp is lit.

When there is a direct current (d.c.) in the primary coil, the lamp is **not** lit.

(i) State two ways in which an alternating current differs from a direct current.

1.

.....

2.

.....

[2]

(ii) Explain why the lamp is **not** lit when there is a direct current in the primary coil.

.....

.....

.....

..... [2]

Mark scheme — 5054/22 Q9 (b), (c)

9(b)(i) transmission of energy B1 (through a medium) with no net movement of medium or by vibrating particles B1 vibrations parallel (and antiparallel) to wave / energy travel direction or cannot be polarised B1 9(b)(ii) 1 two centres of rarefactions labelled R B1 2 distance from one point to adjacent identical point indicated (with double-headed arrow) B1 3. $(v =) f\lambda$ or $25\,000 \times 0.047 / 0.048 / 0.049$ or $25\,000 \times 4.7 / 4.8 / 4.9$ or $25\,000 \times 47 / 48 / 49$ C1 1200 m / s A1 9(c)(i) decreases B1 9(c)(ii) four / five straight lines in air that touch the compressions still in the liquid and no intermediate / extra lines between the correct lines B1 at least four compressions in air parallel to each other B1 at least four straight lines at shallower angle from horizontal and slope correct B1 5054/22 – Mark Scheme PUBLISHED October/November 2018 Page 9 of 10 Question Answer Marks

Question	Answer	Marks
8(c)	any two from: smaller time to drop to zero velocity / hit ground line not straight or velocity does not change uniformly or gradient not constant smaller area under (first part of) graph or less distance travelled slower final velocity initial downward gradient steeper	B2

Question	Answer	Marks
9(a)	frequency of sound wave small(er) or its frequency is less than 20 000 Hz	B1
9(b)(i)	transmission of energy	B1
	(through a medium) with no net movement of medium or by vibrating particles	B1
	vibrations parallel (and antiparallel) to wave / energy travel direction or cannot be polarised	B1
9(b)(ii)	1 two centres of rarefactions labelled <i>R</i>	B1
	2 distance from one point to adjacent identical point indicated (with double-headed arrow)	B1
	3. $(v =) f\lambda$ or $25\,000 \times 0.047 / 0.048 / 0.049$ or $25\,000 \times 4.7 / 4.8 / 4.9$ or $25\,000 \times 47 / 48 / 49$	C1
	1200 m / s	A1
9(c)(i)	decreases	B1
9(c)(ii)	four / five straight lines in air that touch the compressions still in the liquid and no intermediate / extra lines between the correct lines	B1
	at least four compressions in air parallel to each other	B1
	at least four straight lines at shallower angle from horizontal and slope correct	B1

Question	Answer	Marks
9(d)	object (to be cleaned) immersed in liquid / solvent	B1
	object / liquid agitated / vibrated by ultrasound	B1
	dirt (particles) shaken off or dislodges / removes dirt	B1

Question	Answer	Marks
10(a)	electrons c.a.o.	M1
	towards the ammeter or away from the negative terminal or towards the positive terminal	A1
10(b)(i)	thermistor c.a.o.	B1
10(b)(ii)	$1/R_T = 1/R_1 + 1/R_2$ or $1/R_T = 1/1.5 + 1/6.0$ or $(R_T =) R_1 R_2 / (R_1 + R_2)$ or $1.5 \times 6.0 / (1.5 + 6.0)$	C1
	1.2 (Ω)	C1
	2.5 Ω	A1
10(b)(iii)	$(I =) V/R$ or 12 / 2.5	C1
	4.8 (A)	A1
10(b)(iv)	$I_A = I_R + I_Z$	B1
10(c)	resistance of Z / thermistor decreases	B1
	resistance of parallel combination decreases or total resistance (of circuit) decreases or current increases	B1
	voltage (across 1.3 Ω) increases	B1
	trace moves towards top of screen / upwards	B1

Question 10 • included (a)

not included: (b) speed-of-sound experiment

- (a) (i) Determine the distance between the two towns.

distance = [1]

- (ii) Determine the time for which the cyclist rests during the journey, before he reaches town B.

time = [1]

- 2 (a) Some sources of energy, used to produce electricity, may run out in the future.

Underline the **two** sources of energy that are likely to run out first, before the others.

coal oil solar tides wave wind

[1]

- (a) State the form of energy stored in the stretched spring and the form of energy stored in the muscles of the athlete.

form of energy in spring

form of energy in muscles

[2]

- (a) A liquid occupies a much smaller volume than a gas which has the same number of molecules.

Explain why there is this difference.

.....

.....

..... [2]

- (a) Describe how the student uses the apparatus in Fig. 5.1 to obtain a range of ammeter and voltmeter readings.

.....

..... [1]

Mark scheme — 5054/21 Q10 (a)

10(a)(i) 1.7 (cm) seen or any multiple of wavelength measured on diagram C1 6.4–7.2 cm A1 10(a)(ii) number of waves/cycles/oscillations in one second / unit time C1 number of wavelengths/crests/troughs/ wave cycles/wave oscillations generated/made/pass a point in one second/unit time or number of oscillations of a source/particle in one second / unit time A1 10(a)(iii) 2.5 Hz B1 10(a)(iv) moved up and down B1 regularly or at constant frequency / 2.5 times a second B1 10(a)(v) speed smaller / decreases and wavelength smaller / decreases B1

Question	Answer	Marks
10(a)(i)	1.7 (cm) seen or any multiple of wavelength measured on diagram	C1
	6.4–7.2 cm	A1
10(a)(ii)	number of waves/cycles/oscillations in one second / unit time	C1
	number of wavelengths/crests/troughs/ <u>wave</u> cycles/ <u>wave</u> oscillations generated/made/pass a point in one second/unit time	A1
	or number of oscillations of a <u>source/particle</u> in one second / unit time	
10(a)(iii)	2.5 Hz	B1
10(a)(iv)	moved up and down	B1
	regularly or at constant frequency / 2.5 times a second	B1
10(a)(v)	speed smaller / decreases and wavelength smaller / decreases	B1
10(b)(i)	tape measure or trundle wheel or microphones	B1
	stopwatch or timer	B1
10(b)(ii)	start timer on seeing the smoke from gun / sound picked up by microphone	B1
	stop timer on hearing sound	B1
	measure (large) distance between students	B1
10(b)(iii)	both between and including 1000 and 10 000 m/s	B1
	solid speed > liquid speed	B1

Oct/Nov 5054/22

Question 5 • included (a), (b)

not included: (b)(ii) how ultrasound cleans

(a) The energy absorbed by solar panels comes from the Sun.

(i) Describe how this energy is produced inside the Sun.

.....

.....

..... [2]

(ii) State how this energy is transferred to Earth.

..... [1]

(b) Explain why solar panels are usually black.

.....

.....

.....

..... [2]

[Total: 5]

(a) State two ways in which evaporation differs from boiling.

1.

.....

2.

.....

[2]

(b) Describe, in terms of water molecules, what is happening as the water evaporates.

.....

.....

..... [2]

5 Both sound and ultrasound are longitudinal waves.

(a) Describe what is meant by *longitudinal*.

.....

.....

..... [2]

(b) Fig. 5.1 shows an ultrasound cleaner being used to clean jewellery.

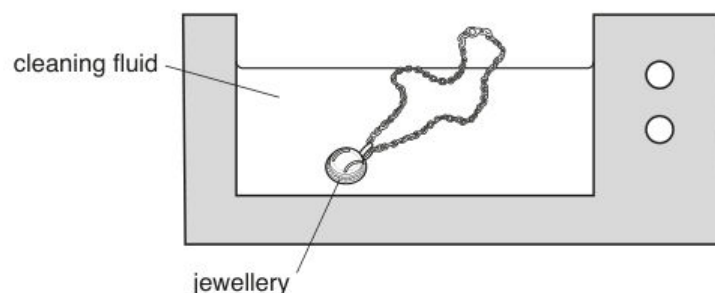


Fig. 5.1

The jewellery is lowered into the cleaning fluid and ultrasound waves of frequency 42 000 Hz are produced in the fluid.

(i) The speed of ultrasound in the fluid is 1500 m/s.

Calculate the wavelength of the ultrasound in the fluid.

wavelength = [2]

(ii) Describe how the ultrasound cleans the jewellery.

.....

.....

..... [2]

[Total: 6]

Mark scheme — 5054/22 Q5 (a), (b)

5(a) vibration / oscillation of particles or energy transfer or compressions and rarefactions B1 vibration / oscillation (of particles) parallel to energy travel direction / propagation direction B1 5054/22 – Mark Scheme PUBLISHED October/November 2019 Page 6 of 10 Question Answer Marks
5(b)(i) ($\lambda =$) v / f or $1500 / 42\,000$ C1 0.036 m or 3.6 cm A1 5(b)(ii) (ultrasound) vibrates (the cleaning fluid / jewellery) B1 dirt shaken off B1
Question Answer Marks

Question	Answer	Marks
3(a)(i)	(nuclear) <u>fusion</u> (reactions) or nuclei fuse	B1
	small <u>nuclei</u> / hydrogen <u>nuclei</u> joining together (and release energy) or larger <u>nucleus</u> produced	B1
3(a)(ii)	infra-red (radiation) or ultraviolet (radiation) or (visible) light	B1
3(b)	black surfaces are (good) absorbers	M1
	of (infra-red) radiation / heat / (visible) light / thermal energy	A1

Question	Answer	Marks
4(a)	any two from: only occurs at the surface / no bubbles produced occurs at any temperature produces cooling	B2
4(b)	molecules escape (from liquid)	B1
	fast(est) molecules / molecules with great(est kinetic) energy escape or slow(est) molecules / molecules with small(est kinetic) energy remain or (average) speed of remaining molecules decreases or molecules gain (kinetic) energy and escape	B1
4(c)(i)	rate of evaporation increases (with increasing area)	B1
4(c)(ii)	rate of evaporation decreases with decreasing temperature or takes too much time	B1

Question	Answer	Marks
5(a)	vibration / oscillation of <u>particles</u> or energy transfer or compressions and rarefactions	B1
	vibration / oscillation (of particles) parallel to energy travel direction / propagation direction	B1

Question	Answer	Marks
5(b)(i)	$(\lambda =) v / f$ or 1500 / 42 000	C1
	0.036 m or 3.6 cm	A1
5(b)(ii)	(ultrasound) vibrates (the cleaning fluid / jewellery)	B1
	dirt shaken off	B1

Question	Answer	Marks
6(a)(i)	4.0 V c.a.o.	B1
6(a)(ii)	$(V =) 12 - 4.0$ or 8.0 or $4.0 / 700$ or 5.7×10^{-3}	C1
	$(R =) V / I$ or $8.0 \times 1 / (4.0 / 700)$ or $700 \times (8.0 / 4.0)$ or $12 / (4.0 / 700)$ or 2100 (Ω)	C1
	1400 Ω	A1
6(b)	resistance of LDR decreases	B1
	current increases or p.d. across 700 Ω resistor / oscilloscope increases or p.d. across LDR decreases	M1
	trace moves up the screen	A1

Question	Answer	Marks
7(a)(i)	current and <u>magnetic field</u> / <u>magnetic flux</u> (experiences a force) or the two <u>magnetic fields</u> interact	B1
7(a)(ii)	three separate approaches I / II / III	
7(a)(iii)	B to A	B1
	left-hand rule mentioned / described	B1
	or	

Question 4 · included (a)

not included: (b) pitch / X-rays / EM-spectrum corrections

- (a) The results are shown in **Fig. 2.2**.

Draw a line linking each type of surface with the appropriate meter reading. One line has been drawn for you.

type of surface	meter reading
dull black	40
dull white	60
shiny black	80
shiny white	100

Fig. 2.2

[2]

- (a) The steel rod has a mass of 2.0 kg and is initially at a temperature of 500 °C. It cools to 50 °C when placed in the water.

The specific heat capacity of steel is 460 J/(kg °C).

Calculate the thermal energy (heat) lost by the steel rod as it cools to 50 °C.

thermal energy = [3]

- 4 (a) Fig. 4.1 shows a wave on a rope and Fig. 4.2 shows a wave on a spring. Both waves are moving in the direction shown by the arrows.



Fig. 4.1

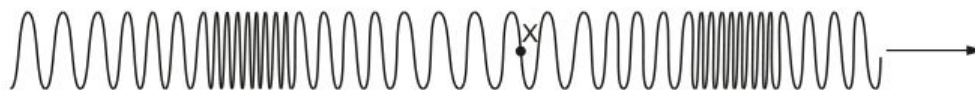


Fig. 4.2

- (i) State the name of each of the two types of wave shown.

Fig. 4.1

Fig. 4.2

[1]

- (ii) On Fig. 4.1, draw a wave which has a smaller frequency than the wave shown. [1]

- (iii) Describe the movement of point X on the spring in Fig. 4.2 as the wave passes.

.....

..... [1]

Mark scheme — 5054/21 Q4 (a)

4(a)(i) Fig.4.1 transverse and Fig.4.2 longitudinal B1 4(a)(ii) wave drawn with larger wavelength B1 4(a)(iii) backwards and forwards, left and right, to and fro, vibration parallel to arrow / direction of wave (movement) B1

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Question	Answer	Marks
3(a)	$(E =) mcT$ in any form algebraic or numerical	C1
	$2 \times 460 \times 450$	C1
	$4.1 \times 10^5 \text{ J}$	A1
3(b)(i)	any one of: - boiling occurs at one temperature / boiling point or evaporation occurs at any temperature / below boiling point - boiling occurs within liquid or evaporation occurs on the surface - evaporation increased by draughts, more surface area (and boiling does not)	B1
3(b)(ii)	(average) speed / K.E. of molecules decreases (as temperature falls) or (molecules) do not have the energy to escape / break bonds	B1
	fewer fast / energetic molecules escape or (only) fastest / high KE molecules escape	B1
Question	Answer	Marks
4(a)(i)	Fig.4.1 transverse and Fig.4.2 longitudinal	B1
4(a)(ii)	wave drawn with larger wavelength	B1
4(a)(iii)	backwards and forwards, left and right, to and fro, vibration parallel to arrow / direction of wave (movement)	B1
4(b)	frequency	B1
	ultrasound	B1
	gamma	B1

Question 10 • included (a)

not included: (b) visible spectrum

- (a) Describe how the reading on the newton meter is used to find the mass of the brick.

.....
 [1]

- 4 (a) In a demonstration, a teacher uses a loud ticking clock, two hollow tubes A and B, a barrier and a smooth surface. The clock is used as a source of sound.

Fig. 4.1 shows tube A and the clock, both fixed in position on the left of the barrier.

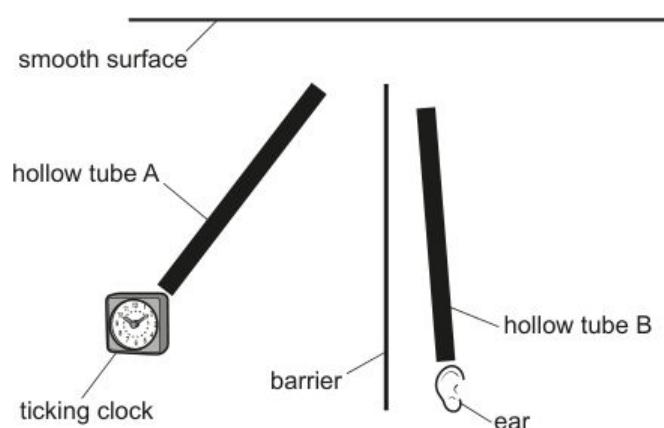


Fig. 4.1

Sound from the ticking clock passes along tube A and is incident on the smooth surface.

A student listens to the sound passing along tube B, which is on the right of the barrier.

- (i) On Fig. 4.1, draw tube B in the position where the sound heard by the student is loudest. [1]

- (ii) Explain your answer to (i).

.....

 [2]

Question 5 begins over the page

- (a) On Fig. 5.2, draw the electric field between the two metal plates. Show the direction of the electric field.

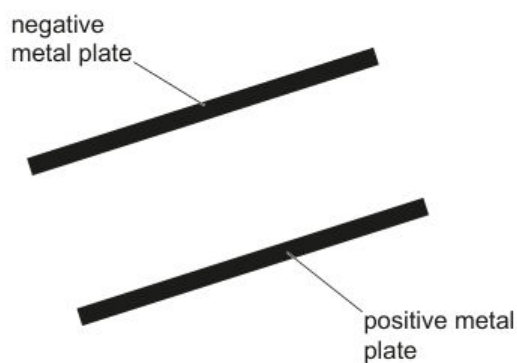


Fig. 5.2

[2]

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Mark scheme — 5054/22 Q10 (a)

10(a)(i) 3 (.0) cm B1 10(a)(ii) number of waves per second or $5 / 10$ or $f = 1 / T$ C1 2(.0) Hz A1 10(a)(iii) $(v =) f \lambda$ or $s = d / t$ numerical or algebraic in any form C1 6(.0) cm / s A1 10(a)(iv) frequency stays the same and speed increases B1 5054/22 – Mark Scheme PUBLISHED May/June 2021
 Page 15 of 16 Question Answer Marks 10(a)(v) ■ three consecutive incident wavefronts meeting straight refracted wavefronts on the boundary without any wavefronts in- between B1 constant wavelength shown between all refracted wavefronts B1 all wavefronts refracted in correct direction, i.e. lines between the vertical and the normal B1

Question	Answer	Marks
9(c)(i)	($E = mc^2$) numerical or algebraic in any form	C1
	$0.11 \times 830 \times 5$	C1
	460 J	A1
9(c)(ii)	energy output / energy input	C1
	useful energy output / (total) energy input	A1
9(c)(iii)	(energy input =) 460 + 5200 or 5660 (J) seen or 5200 ÷ (energy in 9(c)(i) + 5200)	C1
	92% or 0.92	A1
9(c)(iv)	(thermal) energy <u>lost</u> / heat <u>lost</u> (from battery to environment)	B1
	energy input larger than calculated / 5660 (J) or temperature rise should be larger / is smaller than it should be or thermal energy / heat produced larger than 460 J / calculated	B1

Question	Answer	Marks
10(a)(i)	3 (.0) cm	B1
10(a)(ii)	number of waves per second or 5 / 10 or $f = 1 / T$	C1
	2(.0) Hz	A1
10(a)(iii)	($v = f \lambda$ or $s = d / t$) numerical or algebraic in any form	C1
	6(.0) cm / s	A1
10(a)(iv)	frequency stays the same and speed increases	B1

Question	Answer	Marks
10(a)(v)	> three consecutive incident wavefronts meeting straight refracted wavefronts on the boundary without any wavefronts in-between	B1
	constant wavelength shown between all refracted wavefronts	B1
	all wavefronts refracted in correct direction, i.e. lines between the vertical and the normal	B1
10(b)(i)	four colours from violet, indigo, blue, green, yellow, orange, red	B1
	correct order of four of the above colours	B1
10(b)(ii)	prism	B1
	slit (to produce narrow beam)	B1
	dispersion shown at first or second face, i.e. a ray becoming at least two rays at least one face	B1
	correct refraction on first and second faces for all rays shown	B1

Question	Answer	Marks
11(a)	electron	B1
11(b)	prevents exposure to radiation / for safety (of experimenter and others)	B1
	beta particles stopped / absorbed by metal / case or prevents radiation travelling in all directions / to surroundings	B1
11(c)(i)	decreases (slightly) / little change	B1
	particles spread out or <u>some</u> stopped / absorbed (by air)	B1
11(c)(ii)	no particles detected or <u>large</u> decrease	B1
	(>1m) <u>air</u> stops (beta) particles / <u>air</u> atoms / particles become ionised	B1

Question 8 • included (b)

3.1 parts in (b): (iii) long vs trans, (v) $v = f\lambda$ • not motor / loudness

- (b) An uncharged, conducting sphere is suspended from an insulating thread. The positively charged rod is placed near to the sphere, as shown in Fig. 5.1.

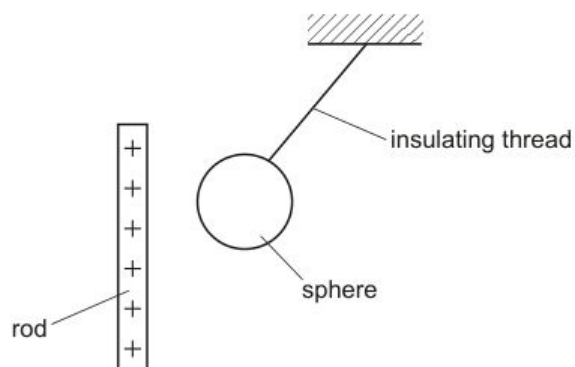


Fig. 5.1

- (i) On Fig. 5.1, draw the distribution of charge on the sphere. [2]
- (ii) Explain why the sphere is pulled towards the rod.

.....

.....

.....

..... [2]

[Total: 6]

(b) The potential difference (p.d.) across the variable resistor is 2.5 V.

(i) State what is meant by *potential difference*.

.....

.....

.....

..... [2]

(ii) Calculate the resistance of the variable resistor.

resistance = [3]

[Total: 6]

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[Turn over

(b) When the magnet is inside the solenoid, it stops moving. It is then pulled back out of the solenoid.

Explain what happens to the ammeter reading as the magnet moves out of the left-hand end of the solenoid.

.....

.....

..... [2]

[Total: 5]

OR

Fig. 7.2 shows the symbol for a logic gate and its truth table.

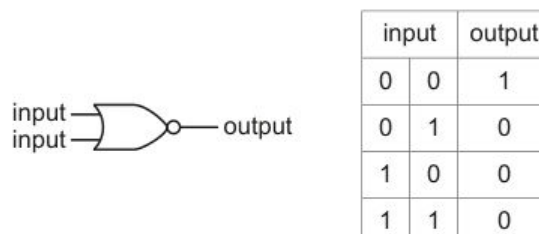


Fig. 7.2

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(b) Fig. 7.3 shows two of these logic gates connected to make a bistable circuit.

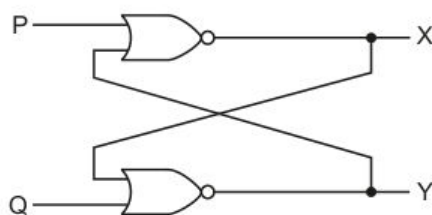


Fig. 7.3

Terminals P and Q each have the logic level 0.

Terminal X has the logic level 1.

(i) Explain why terminal Y has the logic level 0.

.....
 [1]

(ii) Terminal P is briefly connected to a supply of logic level 1.

State the effect of this on the logic level of terminal X and the logic level of terminal Y.

.....
 [1]

(iii) Terminal P returns to logic level 0 again.

State the effect of this return to value 0 on the logic level of terminal X and on the logic level of terminal Y.

.....
 [1]

(iv) State how this circuit exhibits the property of memory.

.....
 [1]

[Total: 5]

- (b) A student connects the coil to the output of an alternating current (a.c.) generator. Fig. 8.2 shows how the electromotive force (e.m.f.) produced by the generator varies with time.

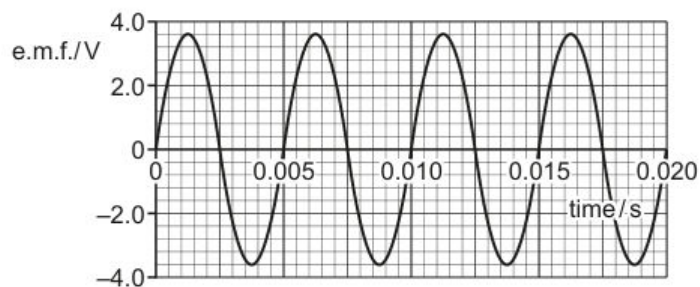


Fig. 8.2

The coil, tube and cone vibrate backwards and forwards.

- (i) Explain why the e.m.f. shown in Fig. 8.2 makes the coil vibrate.

.....

.....

..... [2]

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Mark scheme — 5054/21 Q8 (b)

8(b)(i) the current (in coil continually) reverses (direction) or the force (on the coil continually) reverses B1 the current (in the coil continually) reverses (direction) and the force (on the coil continually) reverses B1 5054/21 – Mark Scheme PUBLISHED October/November 2021 Page 9 of 11
 Question Answer Marks 8(b)(ii) cone moves forwards and air is compressed / hits (air) molecules B1 cone moves backwards and rarefaction produced / molecules sucked back B1 (air) molecules vibrate or production of compressions and rarefactions continues B1 8(b)(iii) vibration direction is parallel to energy travel / propagation direction B1 in transverse wave, vibration direction is perpendicular to propagation / energy travel direction B1 8(b)(iv) (time period =) 0.0050 (s) or 1 / 0.0050 or 200 (times per second) C1 400 (times per second) A1 8(b)(v) (λ =) c / v or 340 / 200 C1 1.7 m A1

Question	Answer	Marks
7E(a)	magnetic field lines cut the solenoid / coils or changing magnetic field in solenoid	B1
	an electromotive force / e.m.f. is <u>induced</u>	B1
	electromotive force / e.m.f. produces the current	B1
7E(b)	negative / opposite / reversed current / deflection / reading	B1
	negative e.m.f. / voltage (induced) or magnetic field cut in opposite direction or magnetic field changes in opposite sense / decreases (owtte)	B1
7O(a)	NOR (gate)	B1
7O(b)(i)	lower logic gate has an input of logic level 1 (and so its output / Y is 0) or the output X has a logic level 1 and so the inputs must both be 0 (Y is an input to the upper NOR gate)	B1
7O(b)(ii)	X changes to 0 and Y changes to 1.	B1
7O(b)(iii)	both X and Y stay at the new values or neither X nor Y changes	B1
7O(b)(iv)	the output shows which terminal (P or Q) last had the value 1.	B1

Question	Answer	Marks
8(a)(i)	magnetic field or magnet	B1
8(b)(i)	the current (in coil continually) reverses (direction) or the force (on the coil continually) reverses	B1
	the current (in the coil continually) reverses (direction) and the force (on the coil continually) reverses	B1

Question	Answer	Marks
8(b)(ii)	cone moves forwards and air is compressed / hits (air) molecules	B1
	cone moves backwards and rarefaction produced / molecules sucked back	B1
	(air) molecules vibrate or production of compressions and rarefactions continues	B1
8(b)(iii)	vibration direction is parallel to energy travel / propagation direction	B1
	in <u>transverse</u> wave, vibration direction is perpendicular to propagation / energy travel direction	B1
8(b)(iv)	(time period =) 0.0050 (s) or 1 / 0.0050 or 200 (times per second)	C1
	400 (times per second)	A1
8(b)(v)	$(\lambda =) c / v$ or 340 / 200	C1
	1.7 m	A1
8(c)(i)	decreases / less loud / quieter	B1
	amplitude of sound / vibration decreases or (maximum) force on coil decreases	B1
8(c)(ii)	(pitch) stays the same / unchanged and no change to frequency / time period	B1

Question	Answer	Marks
9(a)(i)	molecules with similar / slightly smaller packing density and most molecules touching	B1
	random arrangement	B1
9(a)(ii)	molecules (of gas) further apart	B1
	(repulsive) forces between molecules (much) smaller	B1

Question 3 • included (a)

not included: (b) TIR

- (a) State what is meant by a *vector*.

.....
..... [1]

- 3 Fig. 3.1 shows light entering a transparent block.

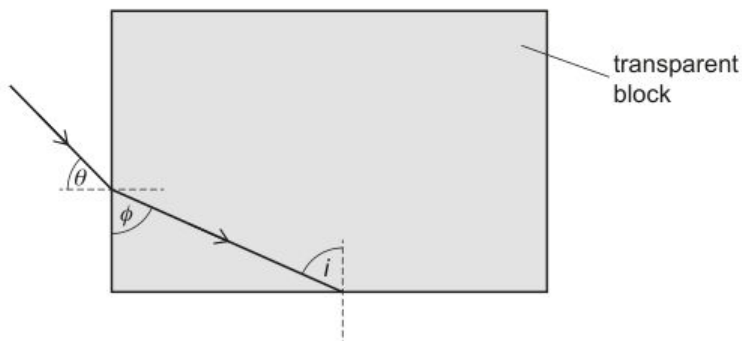


Fig. 3.1 (not to scale)

The light enters the block at an angle θ to the normal and travels through the block until it meets the bottom surface.

The angle between the ray in the block and the vertical side of the block is ϕ .

- (a) Light travels more slowly in the block than in air.

- (i) Explain how Fig. 3.1 shows this.

.....
.....
..... [2]

- (ii) State what happens to the wavelength of the light and what happens to the frequency of the light as it enters the block.

wavelength

frequency

[2]

- (ii) The angle of incidence i at the bottom surface is equal to ϕ and the critical angle for the material of the block in air is 39° .

Explain what happens to the light after it meets the bottom surface.

.....

.....

..... [2]

[Total: 9]

Mark scheme — 5054/22 Q3 (a)

3(a)(i) it / light refracts / bends / moves towards normal or angle of incidence greater than angle of refraction B1 change in direction of the ray / refraction shows a change in speed (at the boundary) or $\sin(i) / \sin(r)$ is ratio of speeds or top of ray slows down first and enters the block first B1
 3(a)(ii) (wavelength) decreases B1 (frequency) does not change. B1

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Question	Answer	Marks
2(b)(ii)	(water) resistance / resistive force / friction (on swimmer)	B1
	(water) resistance / backward force / resistive force / friction / opposing force increases (with speed / time)	B1
	(at constant speed) forward force is equal to resistive / backward force or no resultant force or forces balance	B1

Question	Answer	Marks
3(a)(i)	it / light refracts / bends / moves towards normal or angle of incidence greater than angle of refraction	B1
	change in direction of the ray / refraction shows a change in speed (at the boundary) or $\sin(i) / \sin(r)$ is ratio of speeds or top of ray slows down first and enters the block first	B1
3(a)(ii)	(wavelength) decreases	B1
	(frequency) does not change.	B1
3(b)(i)	$\sin(r) = \sin(i) / n$ or $\sin(45^\circ) / 1.6$ or $0.71 / 1.6$ or 0.44 or $(r =) 21^\circ$	C1
	26°	C1
	64°	A1
3(b)(ii)	64° / angle of incidence / i is greater than the critical angle / 39° or angle of incidence = 64°	B1
	it undergoes total internal reflection	B1

Question	Answer	Marks
4(a)(i)	13 A	B1
4(a)(ii)	0 (A)	B1

2022**May/June 5054/22****Question 8 • included (a), (b), (c)***not included: (d) satellite internet*

- (b) The student connects thermistor X in series with the resistor Y and a battery of e.m.f. 6.0 V, as shown in Fig. 5.2.

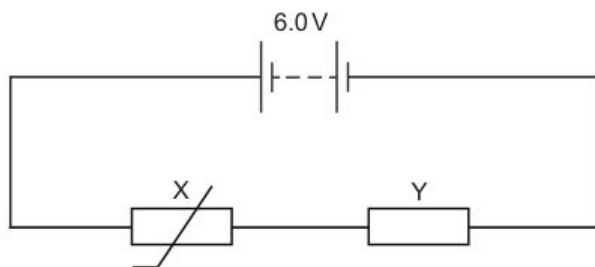


Fig. 5.2

In this circuit, at room temperature, the resistance of thermistor X is $600\ \Omega$ and the current in thermistor X is 0.0020 A.

- (i) Calculate the power produced in the thermistor.

power = [2]

- (ii) Calculate the resistance of Y.

resistance = [3]

- (iii) The thermistor is cooled.

Explain why this causes the potential difference across Y to decrease.

.....

 [2]

[Total: 8]

- 6 (a) Fig. 6.1 shows part of a toy which contains two ring-shaped, permanent magnets. A plastic rod passes through the centre of both magnets.

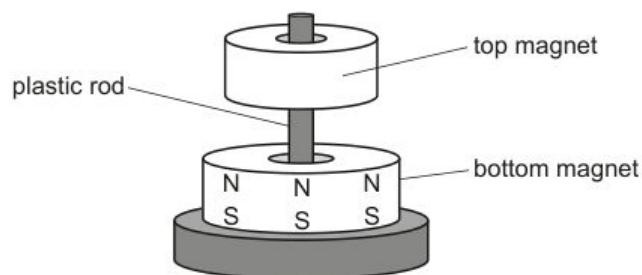


Fig. 6.1

The top magnet can move up and down freely around the plastic rod.

The magnetic poles on the bottom magnet are shown in Fig. 6.1.

- (i) The top magnet floats in the air above the bottom magnet.

On Fig. 6.1, mark the poles on the top magnet and explain why it floats in the air above the bottom magnet.

.....

.....

.....

..... [2]

- (ii) The top magnet is replaced with a ring made of iron.

Explain why the iron ring sticks to the bottom magnet.

.....

.....

.....

..... [2]

- (b) A wire carrying a current passes at right angles through a piece of paper.

Fig. 6.2 shows a cross and circle where the current in the wire passes into the plane of the paper.

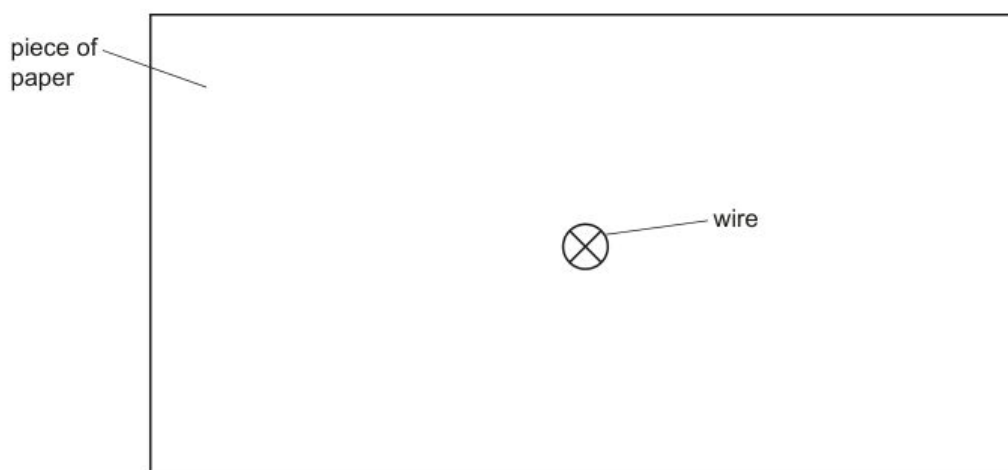


Fig. 6.2

On Fig. 6.2, sketch **three** magnetic field lines to show the magnetic field pattern around the wire.

Show the direction of the field on your sketch.

[2]

[Total: 6]

Section B begins over the page.

- (a) (i) Describe the motion of the car.

.....

.....

.....

.....

..... [3]

- (ii) Using Fig. 7.1, calculate the distance travelled by the car during the 24 s of its motion.

Show your working.

distance = [3]

- (iii) Calculate the average speed of the car during its motion.

average speed = [2]

Mark scheme — 5054/22 Q8 (a), (b), (c)

8(a)(i) 3 B1 8(a)(ii) shape approx. sinusoidal with amplitude 3 mm for the first complete wave (may then change) B1 1 wave taking 0.5 s for first wave shown within half a square or 1.5 s shown for the number of waves in (a)(i) B1 8(b)(i) move bar up and down more often (per second) / faster or make rubber band(s) strong(er) / short(er) / tight(er) / more in number or use bar of less mass / weight B1 8(b)(ii) speed remains the same B1 wavelength decreases B1 8(c)(i) (piece of) plastic / (piece of) glass / block / plate or a shallow(er) / deep(er) area / boundary B1 8(c)(ii) diagram showing within the ripple tank: ■ crests parallel to bar hitting boundary with a boundary labelled B1 ■ correct refraction of plane crests either towards or away from normal on the same side of the normal or no direction change if crests are initially parallel to boundary but with a wavelength change B1 ■ different wavelength shown before and after refraction and constant before and after (by eye) B1

Question	Answer	Marks
8(a)(i)	3	B1
8(a)(ii)	shape approx. sinusoidal with amplitude 3 mm for the first complete wave (may then change)	B1
	1 wave taking 0.5 s for first wave shown within half a square or 1.5 s shown for the number of waves in (a)(i)	B1
8(b)(i)	move bar up and down more often (per second) / faster or make rubber band(s) strong(er) / short(er) / tight(er) / more in number or use bar of less mass / weight	B1
8(b)(ii)	speed remains the same	B1
	wavelength decreases	B1
8(c)(i)	(piece of) plastic / (piece of) glass / block / plate or a shallow(er) / deep(er) area / boundary	B1
8(c)(ii)	diagram showing within the ripple tank:	B1
	crests parallel to bar hitting boundary with a boundary labelled	
	- correct refraction of plane crests either towards or away from normal on the same side of the normal - or no direction change if crests are initially parallel to boundary but with a wavelength change	B1
	different wavelength shown before and after refraction and constant before and after (by eye)	B1
8(d)(i)	($t =$) distance / speed algebraic or numerical seen in any form	C1
	$2 \times 560(000)$ seen or used or 0.0019 (s)	C1
	0.0037 s	A1

Question	Answer	Marks
8(d)(ii)	One of (both): - reflect / refract - have same speed (in a vacuum) - travel in a vacuum / don't need a medium - are electromagnetic - are transverse - carry energy / cause heating - have larger wavelength / smaller frequency than visible light	B1
8(d)(iii)	less likely to be hacked / intercepted / greater security or more channels / better bandwidth possible / more data or less likely for transmission to be interrupted or cost of satellites saved / satellites expensive	B1

Question	Answer	Marks
9(a)(i)	helium atom has (any 1 from) - two electrons (and ion has 1 electron) - one more electron (than the ion) - no charge (and ion positive) - same number of electrons as protons;	B1
9(a)(ii)	alpha particle has (any 1 from) : - no electrons - charge +2 (e) - only protons and neutrons or He ion has an electron	B1

Question 4 · included (b)

only (b)(i) frequency · not EM spectrum / X-ray imaging

- (b)** Calculate the speed of the car when it reaches B.

speed = [3]

(b) The trapped air in the tube is heated.

(i) The height of the trapped air in the tube is x .

Explain, in terms of molecules, why x changes when the air is heated.

.....

.....

.....

.....

..... [3]

(ii) The trapped air reaches a constant temperature that is greater than its initial temperature.

Describe and explain the change in h in terms of the pressures involved.

.....

.....

.....

..... [2]

[Total: 8]

(b) (i) Explain what is meant by 'frequency'.

.....

.....

..... [2]

(ii) State the name of the waves in this spectrum that have the greatest frequency.

..... [1]

Mark scheme — 5054/21 Q4 (b)

4(b)(i) number of oscillations / cycles / wavelengths B1 per unit time B1 4(b)(ii) gamma rays B1

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Cambridge O Level – Mark Scheme
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October/November 2022

Question	Answer	Marks
4(a)	electromagnetic (waves) or components of electromagnetic spectrum	B1
4(b)(i)	number of oscillations / cycles / wavelengths	B1
	per unit time	B1
4(b)(ii)	gamma rays	B1
4(c)(i)	X-rays directed towards body part with (suspected) broken bone	B1
	X-rays pass through flesh but not (to same extent) through bones	B1
	image produced by digital / electronic / CCD / CMOS detector	B1
4(c)(ii)	X-rays (may) cause cancer / cell mutation / birth defect / miscarriage or X-rays show the bone structure rather than foetal development or image would not be clear	B1

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Question 4 · included (b), (c)

- (b) The pile-driver lifts the block from the top of a concrete cylinder, through a height of 0.80 m.

The gravitational field strength g is 10 N/kg.

- (i) Calculate the gravitational potential energy gained by the block.

gravitational potential energy = [2]

- (ii) The block is then dropped from rest onto the top of the concrete cylinder.

Calculate the speed of the block just before it hits the concrete cylinder.

speed = [3]

[Total: 7]

(b) A metal screen surrounding the burning gas is heated by the burning gas until it is red-hot. The hot metal screen warms the man who is standing underneath it.

(i) Describe how thermal energy in the red-hot metal screen is transferred to the man and how it warms him.

.....
.....
.....
.....
..... [3]

(ii) At the top of the heater is a metal hood that has a shiny lower surface.

Explain why this makes the energy transfer from the metal screen more efficient.

.....
.....
..... [2]

(iii) The air temperature decreases and the man puts on a black pullover.

Explain **one** way in which wearing the black pullover helps to keep the man warm.

.....
.....
..... [2]

[Total: 8]

- (b) When a sound wave passes through air or water, the molecules of the air or water move.

Describe the motion of the molecules due to the sound wave.

.....
.....
..... [2]

- (c) Describe what is meant by 'a rarefaction'.

.....
..... [1]

Mark scheme — 5054/22 Q4 (b), (c)

4(b) (they) vibrate M1 (vibration) parallel to / in the direction of propagation / of energy travel A1 4(c) (point where) molecules are further apart (than average) or (point where) pressure / density is lower (than average) B1

Question	Answer	Marks
3(a)	chemical potential energy and no other energy	B1
3(b)(i)	radiation or infrared mentioned	B1
	radiation and infrared (travel to the man)	B1
	energy / infrared / radiation <u>absorbed</u> by the man / pullover or travels in straight line or travels at the speed of light or increases (man's) internal energy	B1
3(b)(ii)	(shiny surface) <u>reflects</u> radiation (to man)	B1
	less (thermal) energy escapes or more (thermal) energy reaches / absorbed by man	B1
3(b)(iii)	EITHER black (surface) is good absorber / poor reflector of infrared radiation	B1
	more (thermal) energy absorbed or more thermal energy transferred to man	B1
	OR trapped air and reduced convection / air poor conductor / air is a good insulator	(B1)
	less (thermal) energy lost	(B1)

Question	Answer	Marks
4(a)	equipment e.g. (evacuatable) container and (suspended electric) bell / source of sound and (air) <u>pump</u>	B1
	action e.g. ringing / sound (produced) / sound heard and evacuate air / vacuum (produced)	B1
	observation e.g. transmission / hearing of sound ceases / no sound	B1

Question	Answer	Marks
4(b)	(they) vibrate	M1
	(vibration) parallel to / in the direction of propagation / of energy travel	A1
4(c)	(point where) molecules are further apart (than average) or (point where) pressure / density is lower (than average)	B1
4(d)	$(\lambda =) v / f$ or 1500 / 140 000	C1
	1.1×10^{-2} m or 1.1 cm or 11 mm	A1

Question	Answer	Marks
5(a)	energy / work done per unit charge or $\frac{\text{energy}}{\text{charge}}$ or $\frac{\text{work done}}{\text{charge}}$	B1
	property of a source / cell / battery / power supply or energy used driving charge around a (complete) circuit or energy transferred to electrical energy	B1
5(b)	it lasts for a longer time or more (electrical) energy available or power supply still functions if one cell fails / runs flat or can replace one cell without stopping circuit / stopping current	B1
5(c)(i)	resistance (of thermistor / circuit) decreases c.a.o.	B1
	current change or voltage change across thermistor matching the change in the resistance of the resistor	B1
	reading of voltmeter increases c.a.o.	B1
5(c)(ii)	$(V_1 =) V_0 \times R_1 / R_0$ or $1.5 \times 4000 / (4000 + 8000)$ or $1.5 \times 4000 / 12\,000$ or $I = V / R$ or $1.5 \times 8000 / (4000 + 8000)$ or $I = 1.5 / 12\,000$ or 0.125 mA	C1
	0.50 V	A1

Question 7 · included (a), (b), (c)

(a) Describe the motion of the glider in the first 12 s.

.....

.....

..... [2]

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(b) In the first 6.0 s of the motion, there is a resultant force of 1800 N on the glider.

Using the increase in speed in the first 6.0 s, calculate the mass of the glider.

mass = kg [3]

(c) Determine the distance travelled by the glider in the first 6.0 s of its motion.

distance = m [2]

(a) (i) Define 'momentum'.

.....
..... [1]

(ii) State the unit of momentum.

..... [1]

(b) Calculate the velocity v of the car and van after the collision.

$v =$ m/s [2]

(c) (i) Calculate the total kinetic energy of the car and van after the collision.

kinetic energy = J [2]

(ii) State the transfer of energy that occurs in the collision.

.....
..... [1]

[Total: 7]

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(a) Calculate the current in the $400\ \Omega$ resistor.

current = A [2]

(b) Calculate the power produced in the $400\ \Omega$ resistor.

power = W [2]

(c) A student uses all of the components shown in Fig. 3.1 in another circuit.

She connects them so that there is the largest possible current in each resistor.

The current in one of the resistors is larger than the current in the other two resistors. The student connects the ammeter into the circuit to measure the current in this resistor.

Draw the circuit diagram of the arrangement. Label each resistor with the value of its resistance.

[2]

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[Total: 6]
[Turn over

- 4 (a) A student pushes a drawing pin into a wooden board, as shown in Fig. 4.1.

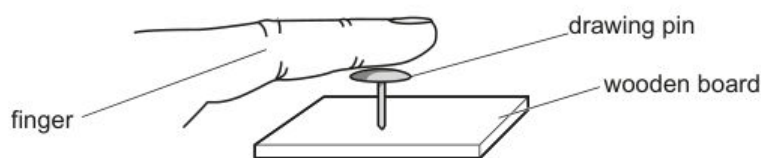


Fig. 4.1

The area of the pin in contact with the finger is $5.0 \times 10^{-5} \text{ m}^2$. The student pushes downwards with a force of 26 N.

The mass of the drawing pin is very small.

- (i) Calculate the pressure exerted by the finger on the drawing pin.

pressure = Pa [2]

- (ii) Compare the force exerted by the finger on the drawing pin with the force exerted by the drawing pin on the wooden board.

..... [1]

- (iii) Explain why the drawing pin goes into the wooden board but not into the finger.

..... [2]

Mark scheme — 5054/21 Q7 (a), (b), (c)

7(a) wavelength has altered / too large or crests in shadow should be bent / quarter circular above or below gap or vertical lines too straight on two crests B1 7(b) at least 3 semicircles centered on the gap B1 wavelength constant and the same on right and left of gap and some bending B1 7(c)(i) distance between two (consecutive) identical points such as two (consecutive) crests B1 7(c)(ii) $v = f \lambda$ numerical or algebraic e.g. $f = 6 / 2$ (Hz) C1 3(.0) Hz) A1 7(c)(iii) 1.3 (cm) B1 5054/21 – Mark Scheme PUBLISHED May/June 2023 Page 11 of 13 Question Answer Marks

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Question	Answer	Marks
7(a)	wavelength has altered / too large or crests in shadow should be bent / quarter circular above or below gap or vertical lines too straight on two crests	B1
7(b)	at least 3 semicircles centered on the gap	B1
	wavelength constant and the same on right and left of gap and some bending	B1
7(c)(i)	distance between two (consecutive) identical points such as two (consecutive) crests	B1
7(c)(ii)	$v = f\lambda$ numerical or algebraic e.g. $f = 6 / 2$ (Hz)	C1
	3(.0) Hz	A1
7(c)(iii)	1.3 (cm)	B1

May/June 5054/22**Question 5 · included (a), (b)**

- (a) (i) Describe why rotating the tube changes the pressure of the gas in the sealed end.

.....

.....

..... [1]

- (ii) Explain, using ideas about particles, why the pressure of the gas decreases when its volume increases.

.....

.....

.....

..... [3]

- (b) In Fig. 3.1 the length of the mercury column is 0.30 m.

The density of mercury is $14\,000\text{ kg/m}^3$.

Atmospheric pressure is $1.0 \times 10^5\text{ Pa}$.

Calculate the pressure of the gas in the tube.

pressure = Pa [3]

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- (a) (i) State the lowest possible temperature on the Celsius scale and on the Kelvin scale.

Celsius scale °C Kelvin scale K [1]

- (ii) The temperature of the solid increases. The sample remains a solid.

Describe how the motion of the particles changes.

.....
..... [2]

- (iii) The solid melts.

State what happens to the internal energy and the temperature of the solid as it melts.

internal energy
temperature [2]

- (b) A student:

- places a 300 g piece of iron in boiling water until the iron is at a temperature of 100°C
- removes the iron from the water and places it immediately into 100 g of water at 25°C .

The iron cools and the water warms until both reach the same temperature, 44°C .

The specific heat capacity of water is $4.2\text{ J/(g}^\circ\text{C)}$. No energy is lost to the surroundings.

- (i) Calculate the change in energy (internal energy) of the water as it warms up.

change in energy = J [2]

- (ii) Calculate the specific heat capacity of iron.

specific heat capacity = $\text{J/(g}^\circ\text{C)}$ [2]
[Total: 9]

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5 Water waves are transverse waves.

(a) Underline **two** other examples of transverse waves.

seismic P-waves seismic S-waves sound X-rays [1]

(b) Fig. 5.1 shows a wooden bar and a glass block in a ripple tank. The depth of water in the tank is less than the height of the glass block.

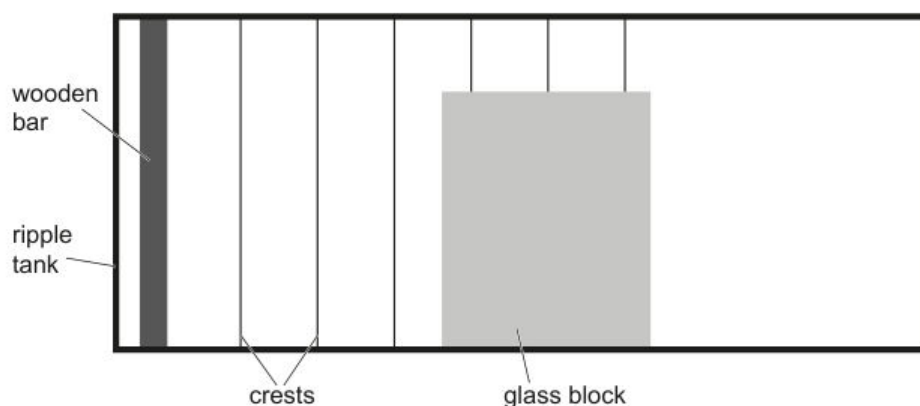


Fig. 5.1 (not to scale)

The wooden bar moves up and down once every 0.15 s to create the crests.

(i) The speed of the water wave is 27 cm/s.

Calculate the frequency and the wavelength of the wave.

frequency = Hz

wavelength = cm
[3]

(ii) The wave diffracts at the right-hand edge of the glass block.

On Fig. 5.1 draw **two** crests after they pass the glass block to show the diffraction. [2]

(iii) Describe how a wave with a smaller wavelength is made with the wooden bar.

.....
..... [1]

(iv) Describe how a decrease in wavelength affects the diffraction.

.....
..... [1]

[Total: 8]

[Turn over]

Mark scheme — 5054/22 Q5 (a), (b)

5(a) seismic S-waves and X-rays B1 5(b)(i) (wavelength =) v / f in any form or (distance =) vt or $f = 1 / T$ or $1 / 0.15$ C1 frequency 6.7 (Hz) A1 wavelength 4.0 (cm) A1 5(b)(ii) straight wavefronts above block to the right with same wavelength B1 correct diffraction – wavefronts quarter circles with centre at top corner B1 5(b)(iii) increase frequency of movement of (wooden) bar B1 5054/22 – Mark Scheme PUBLISHED May/June 2023 Page 9 of 12 Question Answer Marks 5(b)(iv) less diffraction or less bending B1 Question Answer Marks

Question	Answer	Marks
4(a)(i)	–273 (°C) and 0 (K)	B1
4(a)(ii)	particles) move fast(er)	B1
	vibrate / oscillate with larger amplitude	B1
4(a)(iii)	internal energy increases	B1
	temperature stays the same	B1
4(b)(i)	($E = mcT$) in any form numerical or algebraic	C1
	8000 (J)	A1
4(b)(ii)	100–44 or 56	C1
	0.48 or 0.47 (J/(g °C))	A1

Question	Answer	Marks
5(a)	seismic S-waves and X-rays	B1
5(b)(i)	(wavelength =) v / f in any form or (distance =) vt or $f = 1 / T$ or $1 / 0.15$	C1
	frequency 6.7 (Hz)	A1
	wavelength 4.0 (cm)	A1
5(b)(ii)	<u>straight</u> wavefronts above block to the right with same wavelength	B1
	correct diffraction – wavefronts quarter circles with centre at top corner	B1
5(b)(iii)	increase frequency of movement of (wooden) bar	B1

Question	Answer	Marks
5(b)(iv)	less diffraction or less bending	B1

Question	Answer	Marks
6(a)(i)	ammeter A_2 and A_3 both 0.25 (A)	B1
	voltmeter V_2 9(.0 V)	B1
6(a)(ii)	($R =$) V / I in any form	C1
	36 (Ω)	A1
6(a)(iii)	current is (directly) proportional to voltage / potential difference	B1
	provided the temperature is constant	B1
6(b)	0.35 (A) seen	B1
	or total resistance $36 + 6 / 0.35$; $36 + 17(.1)$; $53(.1) \Omega$	
	voltage (across R) 0.35×36 or 12.6 (V) seen	B1
	19 V	B1

Question	Answer	Marks
7(a)(i)	X-rays and gamma (radiation / rays)	B1
7(a)(ii)	security marking / detecting counterfeit bank notes / sterilising water / fluorescent effects (e.g. fingerprints, blood, biological fluids) / (sun) tanning / sun beds / cancer treatment / curing inks and resins (e.g. in dental fillings) / production of vitamin (D) / generating electricity / solar panels / disinfection / sterilisation	B1
7(a)(iii)	(skin) cancer / cataracts / burns or destroys the retina or cornea	B1

Question 4 • included (a), (b)

not included: (b)(ii) refractive index; (b)(iii) ray path

- (a) Define what is meant by 'acceleration'.

.....

.....

..... [2]

- (b) As the lorry moves forwards, work is done on it by the driving force.

- (i) State the equation that defines work done.

Refer to the direction of the force exerted.

.....

..... [1]

- (ii) Calculate the work done on the lorry by the driving force when the lorry travels a distance of 280 m.

work done = J [2]

- (iii) At one point on the road, the kinetic energy of the lorry is $3.2 \times 10^6 \text{ J}$ and its speed is 12 m/s.

Calculate the mass of the lorry.

mass = kg [3]

[Total: 8]

- (a) Explain, by referring to particles, why a liquid expands when heated.

.....
.....
.....
..... [2]

- (b) The heater in a kettle is near to the base of the kettle. The kettle is filled with water at a temperature of 17 °C and the heater is switched on.

- (i) State the name of the method of thermal energy transfer that causes energy to be transferred to the water that is touching the heater.

..... [1]

- (ii) Explain how thermal energy is transferred through all of the water in the kettle.

.....
.....
.....
..... [3]

- (iii) State the boiling temperature of water at standard atmospheric pressure. Include the unit in your answer.

..... [1]

- (iv) The kettle contains 2.5 kg of water at standard atmospheric pressure.

The specific heat capacity of water is 4200 J/(kg °C).

When the water reaches its boiling temperature, the kettle switches off.

Calculate the increase in the internal energy of the water.

increase in internal energy = J [3]

[Total: 10]

- 4 Red light of frequency 4.7×10^{14} Hz travels in air at a speed of 3.0×10^8 m/s.

(a) Calculate the wavelength in air of this red light.

wavelength = m [3]

(b) Fig. 4.1 shows a narrow beam of this red light striking the surface of a parallel-sided glass block.

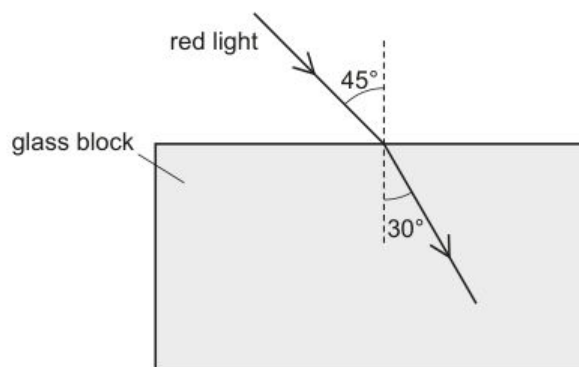


Fig. 4.1

(i) State what happens to the speed, to the frequency and to the wavelength of the light as it enters the block.

speed

frequency

wavelength [2]

(ii) Using the angles shown on Fig. 4.1, calculate the refractive index of the glass.

refractive index = [3]

- (iii) The light continues along the path shown in Fig. 4.1 until it strikes the bottom surface of the block. Light then emerges into the air.

Draw on Fig. 4.1 to show the path taken by the light until it strikes the bottom surface and the path of the light that emerges into the air. [2]

[Total: 10]

Mark scheme — 5054/21 Q4 (a), (b)

4(a) (■ =) v / f or 3.0 ■ 108 / 4.7 ■ 1014 C1 6.4 ■ 10N (m) or 3 / 4.7 ■ 106 C1 6.4 ■ 10^{-7} (m) A1 4(b)(i) speed: decreases frequency: stays the same wavelength; decreases B2 4(b)(ii) $i = 45^\circ$ and $r = 30^\circ$ C1 ($n \Rightarrow \sin(i) / \sin(r)$ or $\sin(45^\circ) / \sin(30^\circ)$) C1 1.4 A1 4(b)(iii) straight continuation of ray in block and refraction away from normal in air B1 emergent ray parallel to initial incident ray B1 Question Answer Marks

Question	Answer	Marks
4(a)	$(\lambda =) v / f$ or $3.0 \times 10^8 / 4.7 \times 10^{14}$	C1
	6.4×10^N (m) or $3 / 4.7 \times 10^6$	C1
	6.4×10^{-7} (m)	A1
4(b)(i)	speed: decreases frequency: stays the same wavelength: decreases	B2
4(b)(ii)	$i = 45^\circ$ and $r = 30^\circ$	C1
	$(n =) \sin(i) / \sin(r)$ or $\sin(45^\circ) / \sin(30^\circ)$	C1
	1.4	A1
4(b)(iii)	straight continuation of ray in block and refraction away from normal in air	B1
	emergent ray parallel to initial incident ray	B1

Question	Answer	Marks
5(a)(i)	infrared (radiation) and visible (light) and ultraviolet (radiation)	B1
	infrared (radiation) and visible (light) and ultraviolet (radiation) and in this order	B1
5(a)(ii)	damaging effect: (skin) cancer / cataracts	B1
	property: it / ultraviolet radiation is ionising	B1

Question 5 · included (b)*only (b) trans vs long · not EM regions*

(b) The heater in a kettle is near to the base of the kettle. The kettle is filled with water at a temperature of 17°C and the heater is switched on.

(i) State the name of the method of thermal energy transfer that causes energy to be transferred to the water that is touching the heater.

..... [1]

(ii) Explain how thermal energy is transferred through all of the water in the kettle.

.....
.....
.....
..... [3]

(iii) State the boiling temperature of water at standard atmospheric pressure. Include the unit in your answer.

..... [1]

(iv) The kettle contains 2.5 kg of water at standard atmospheric pressure.

The specific heat capacity of water is $4200\text{ J}/(\text{kg }^{\circ}\text{C})$.

When the water reaches its boiling temperature, the kettle switches off.

Calculate the increase in the internal energy of the water.

increase in internal energy = J [3]

[Total: 10]

- (b) Fig. 4.1 shows a narrow beam of this red light striking the surface of a parallel-sided glass block.

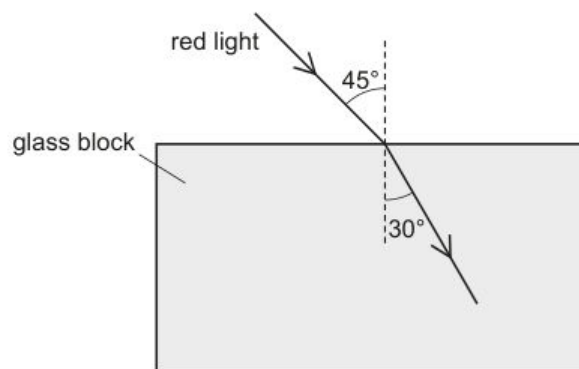


Fig. 4.1

- (i) State what happens to the speed, to the frequency and to the wavelength of the light as it enters the block.

speed

frequency

wavelength

[2]

- (ii) Using the angles shown on Fig. 4.1, calculate the refractive index of the glass.

refractive index = [3]

- (iii) The light continues along the path shown in Fig. 4.1 until it strikes the bottom surface of the block. Light then emerges into the air.

Draw on Fig. 4.1 to show the path taken by the light until it strikes the bottom surface and the path of the light that emerges into the air. [2]

[Total: 10]

(b) Electromagnetic radiation is a transverse wave.

Describe the difference between a transverse wave and a longitudinal wave.

.....

.....

..... [2]

[Total: 6]

Mark scheme — 5054/21 Q5 (b)

5(b) in a transverse wave, the vibration direction is perpendicular / at right angles to the propagation direction B1 in a longitudinal wave, the vibration direction is parallel to the propagation direction B1 Question Answer Marks

Question	Answer	Marks
4(a)	$(\lambda =) v / f$ or $3.0 \times 10^8 / 4.7 \times 10^{14}$	C1
	6.4×10^N (m) or $3 / 4.7 \times 10^6$	C1
	6.4×10^{-7} (m)	A1
4(b)(i)	speed: decreases frequency: stays the same wavelength: decreases	B2
4(b)(ii)	$i = 45^\circ$ and $r = 30^\circ$	C1
	$(n =) \sin(i) / \sin(r)$ or $\sin(45^\circ) / \sin(30^\circ)$	C1
	1.4	A1
4(b)(iii)	straight continuation of ray in block and refraction away from normal in air	B1
	emergent ray parallel to initial incident ray	B1

Question	Answer	Marks
5(a)(i)	infrared (radiation) and visible (light) and ultraviolet (radiation)	B1
	infrared (radiation) and visible (light) and ultraviolet (radiation) and in this order	B1
5(a)(ii)	damaging effect: (skin) cancer / cataracts	B1
	property: it / ultraviolet radiation is ionising	B1

Question	Answer	Marks
5(b)	in a transverse wave, the <u>vibration</u> direction is perpendicular / at right angles to the propagation direction	B1
	in a longitudinal wave, the <u>vibration</u> direction is parallel to the propagation direction	B1

Question	Answer	Marks
6(a)	(because) the current at every point in a series circuit is the same	B1
6(b)(i)	$(R =) V / I$ or $1.5 / 0.20$ or $7.5 (\Omega)$	C1
	$(R_T =) 7.5 - 1.5 = 3.5$	C1
	$2.5 (\Omega)$	A1
6(b)(ii)	resistance (of T / thermistor) decreases (as the temperature increases)	B1
	current in circuit increases or smaller fraction of total resistance	B1
	p.d. across R and S increases or (p.d. is) smaller fraction of e.m.f.	B1

Question	Answer	Marks
7(a)(i)	P: (carbon) brush and Q: slip ring(s)	B1
7(a)(ii)	to connect the output terminal(s) and the (rotating) coil/brushes	B1
	without wires twisting	B1
7(b)(i)	at least one of the five positions when the output e.m.f. is zero indicated with X and no indications elsewhere	B1

Question 4 · included (a), (b), (c)

not included: (a)(iii) use of ultrasound

(a) (i) On Fig. 1.2, mark:

- with the letter P **one** point where the ball accelerates
- with the letter Q **one** point where the ball has constant speed.

[2]

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(b) The speeds of the ball at A, B, C and D are v_A , v_B , v_C and v_D respectively.

Arrange these four speeds from slowest to fastest.

slowest $\longrightarrow$ fastest

--	--	--	--

[1]

(c) Work is done to transfer energy between energy stores as the ball moves from A to B.

Name the force involved in the work done and describe the energy transfer.

.....

.....

.....

..... [3]

- (a) (i) Explain, using ideas about particles, how the steam creates a pressure on the piston.

.....
.....
.....
..... [2]

- (ii) Explain why the pressure on the piston increases as the temperature of the steam increases.

.....
.....
.....
..... [2]

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- (b) When the pressure is high enough, the holdback bar is released. The steam pushes the piston along the pipe shown in Fig. 2.1.

The piston has a cross-sectional area of 0.30 m^2 .

The pressure of the steam in the tank is $2.1 \times 10^6 \text{ Pa}$.

Atmospheric pressure is $1.0 \times 10^5 \text{ Pa}$.

Determine the resultant force on the piston caused by the pressure difference.

Show your working.

force = N [2]

- (c) The force calculated in (b) causes the aircraft to accelerate to a maximum speed of 28 m/s from rest.

The mass of the aircraft is 3.0×10^4 kg.

- (i) Calculate the momentum of the aircraft at the maximum speed.

momentum = kg m/s [2]

- (ii) Calculate the time taken for the aircraft to reach the maximum speed.

time = s [2]

[Total: 10]

- (a) Thermal conduction occurs in the metal from which the pan is made.

By referring to particles, describe the process of thermal conduction in a metal.

.....
.....
.....
..... [2]

(b) Fig. 3.2 shows how the temperature of the contents of the pan varies with time.

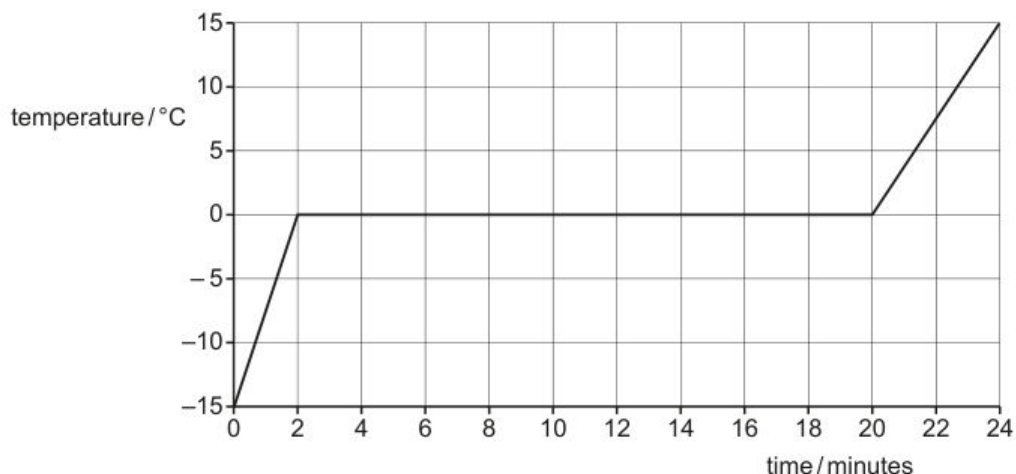


Fig. 3.2

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Mark scheme — 5054/21 Q4 (a), (b), (c)

4(a)(i) number of (complete) waves (passing a point) per second B1 4(a)(ii) ($V =$) $f\lambda$ in any form C1 330 m / s A1 4(a)(iii) ANY one from cleaning, prenatal and other medical scanning, sonar, calculation of depth or distance B1 4(b)(i) longitudinal wave - oscillation backwards and forwards / in direction of travel of wave B1 transverse wave - oscillation at right angles to (direction of travel of) wave B1 diagram of each type B1 4(b)(ii) earthquake (waves) or other correct source which is a single source that generates both types of wave, e.g. lightning B1 4(c) long wavelengths bend around hill and short wavelengths do not bend B1 (due to) diffraction B1 Question Answer Marks

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Question	Answer	Marks
3(b)(v)	(liquid) water has a higher specific heat capacity; water has stronger bonds	B1

Question	Answer	Marks
4(a)(i)	number of (complete) waves (passing a point) per second	B1
4(a)(ii)	$(V =) f \lambda$ in any form	C1
	330 m / s	A1
4(a)(iii)	ANY one from cleaning, prenatal and other medical scanning, sonar, calculation of depth or distance	B1
4(b)(i)	longitudinal wave - oscillation backwards and forwards / in direction of travel of wave	B1
	transverse wave - oscillation at right angles to (direction of travel of) wave	B1
	diagram of each type	B1
4(b)(ii)	earthquake (waves) or other correct source which is a single source that generates both types of wave, e.g. lightning	B1
4(c)	long wavelengths bend around hill and short wavelengths do not bend	B1
	(due to) diffraction	B1

Question	Answer	Marks
5(a)	correct symbol; 	B1
5(b)(i)	correct curvature	B1
5(b)(ii)	(at higher current / voltage) temperature increases	B1
	(at higher current / voltage) resistance increases	B1

Oct/Nov 5054/21
Question 5 • included (a), (b), (c)

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5054/21/O/N/24

[Turn over]

- (a) Explain, in terms of electrons and the atoms of the metal, how thermal energy is transferred through the walls of the metal cylinder.

.....

.....

.....

.....

.....

..... [3]

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5054/21/O/N/24

- (b) The piston is fixed in position so the pressure of the gas increases as the temperature of the gas increases.

- (i) State what happens to the motion of the particles of the gas as the temperature increases.

.....

..... [1]

- (ii) Explain, in terms of the particles of the gas, why the pressure the gas exerts on the walls of the metal cylinder increases as the temperature increases.

.....

.....

.....

.....

..... [3]

- (c) The gas in the cylinder reaches a temperature of 100°C .

The piston is released and moves to the right. The temperature remains at 100°C .

The pressure of the gas now decreases.

- (i) Explain, in terms of the particles of the gas, why the pressure of the gas now decreases.

.....

 [1]

- (ii) Eventually, the piston stops moving.

Explain why.

.....

 [2]

[Total: 10]



- (a) Draw on Fig. 4.1 to show the paths of the red light and the violet light between the left-hand side of the prism and the screen at point R and point V. [2]

- (b) State what these observations show about the change in speed of red light and change in speed of violet light as each enters the glass from the air.

- (b) State what these observations show about the change in speed of red light and change in speed of violet light as each enters the glass from the air.

.....

 [2]

- (c) Five other colours of light appear on the screen between the red light at point R and the violet light at point V.

- (i) State the names of the five colours and list them in the correct order from red to violet.

red violet [2]

- (ii) State how the frequency and the wavelength of the coloured light change going from red light to violet light.

frequency
 wavelength [1]

- 5 Ultrasound is a longitudinal wave which cannot be heard by humans as the frequency of ultrasound is too high.

- (a) Ultrasound waves **cannot** travel in a vacuum.

Explain why.

.....

 [2]

- (b) Describe how a longitudinal wave differs from a transverse wave.

.....

 [2]

- (c) An ultrasound transmitter used for a medical scan produces an ultrasound wave of frequency 8.4 MHz.

In soft human tissue, ultrasound travels at 1500 m/s.

- (i) Calculate the wavelength of this ultrasound wave in soft human tissue.

wavelength = m [3]

- (ii) The ultrasound wave passes from the soft tissue into bone where the speed of the ultrasound wave is greater than 1500 m/s.

State what happens to the frequency and to the wavelength of the ultrasound wave as it passes into the bone.

frequency

wavelength

[2]

[Total: 9]



Mark scheme — 5054/21 Q5 (a), (b), (c)

5(a) ultrasound / sound / it is transmitted by vibrating matter / particles B1 no matter / particles in a vacuum / no medium B1 5(b) (in a longitudinal wave,) vibration (of particles) is parallel to the propagation direction B1 in a transverse wave, vibration (of particles) is perpendicular to the propagation direction B1 5(c)(i) (■ =) v / f or $1500 / 8.4$ ■ 106 C1 1.8 ■ 10N C1 1.8 ■ 10–4 (m) A1 5(c)(ii) frequency: does not change B1 wavelength: increases B1 5054/21 – Mark Scheme PUBLISHED October/November 2024 © Cambridge University Press & Assessment 2024 Page 8 of 10 Question Answer Marks

Question	Answer	Marks
4(c)(ii)	frequency increases and wavelength decreases	B1
4(d)	infrared (radiation)	B1

Question	Answer	Marks
5(a)	ultrasound / sound / it is transmitted by <u>vibrating</u> matter / particles	B1
	no matter / particles in a vacuum / no medium	B1
5(b)	(in a longitudinal wave,) vibration (of particles) is parallel to the propagation direction	B1
	in a transverse wave, vibration (of particles) is perpendicular to the propagation direction	B1
5(c)(i)	$(\lambda =) v / f$ or $1500 / 8.4 \times 10^6$	C1
	1.8×10^N	C1
	1.8×10^{-4} (m)	A1
5(c)(ii)	frequency: does not change	B1
	wavelength: increases	B1

2025
May/June 5054/21**Question 5 · included (a), (b), (c)**

- (c) A tall bus is tested for stability.

Fig. 2.2 shows the bus on a slope.

The centre of gravity of the bus is marked.

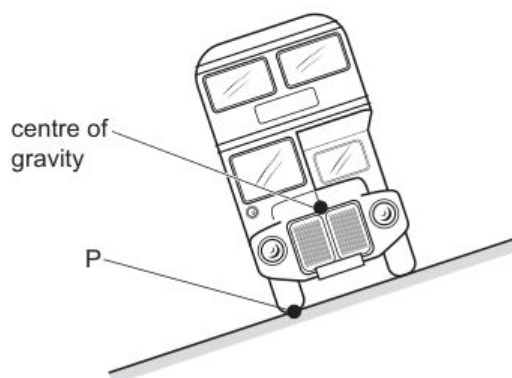


Fig. 2.2

- (i) State what is meant by the 'centre of gravity' of an object.

.....
 [1]

- (ii) When the slope is made very steep, the bus falls over by rotating about point P.

Explain why the bus falls over.

.....

 [2]

[Total: 8]



- (a) Describe the transfers between energy stores that occur as the load is lifted.

.....

.....

.....

..... [3]

- (b) The efficiency of the motor, pulley and load system is less than 100%.

- (i) By comparing the input energy and the useful output energy, explain why the efficiency is less than 100%.

.....

..... [1]

- (ii) Explain how the principle of conservation of energy applies in lifting the load.

.....

.....

..... [2]

- (c) The input power to the motor is 15 W. The motor is used for 20 s. The efficiency of the motor is 60%.

Calculate the energy supplied to the load.

energy = J [2]

[Total: 8]

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5054/21/M/J/25

- 4 (a) Evaporation of water from the surface of the skin causes cooling.

- (i) Describe, using ideas about particles, how evaporation causes cooling.

.....

.....

..... [2]

- (ii) State **one** difference between evaporation and boiling.

.....

..... [1]

- (b) When a refrigerator is switched on, cooling coils placed at the top of the space inside the refrigerator become cold. This causes a convection current which cools the air inside the refrigerator. The refrigerator is shown in Fig. 4.1.

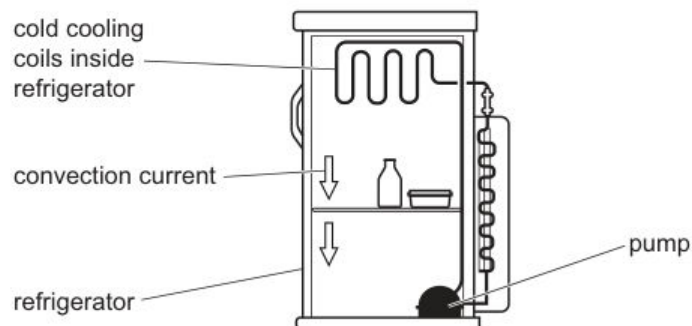


Fig. 4.1

- (i) Explain how the cooling coils cause a convection current in the air inside the refrigerator.

.....

 [2]

- (ii) The food in the refrigerator is initially at a temperature of 20.0°C . The food has a mass of 3.6 kg and a specific heat capacity of $3000\text{ J}/(\text{kg }^{\circ}\text{C})$.

Calculate the final temperature of the food after $160\,000\text{ J}$ of thermal energy is removed from it.

final temperature = $^{\circ}\text{C}$ [2]

[Total: 7]

[Turn over]

- 5 (a) Fig. 5.1 is a diagram showing the arrangement of air particles as a longitudinal wave passes through them.



Fig. 5.1

- (i) On Fig. 5.1, mark the centre of a compression with the letter C, and mark the centre of a rarefaction with the letter R. [1]

- (ii) Describe the difference between a compression and a rarefaction.

.....

 [1]

(b) In a ripple tank, a water wave is produced by a wooden bar moving up and down on the surface of water.

(i) The wooden bar makes 45 complete oscillations in 1.0 minute.

Calculate the frequency of the wave produced.

frequency =Hz [1]

(ii) The frequency of the water wave is increased by moving the wooden bar up and down more quickly.

State what happens to the speed and what happens to the wavelength of the wave produced.

speed

wavelength [2]

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5054/21/M/J/25

(c) Describe what is meant by the diffraction of a water wave.

.....
.....
..... [1]

[Total: 8]

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[Turn over]

Mark scheme — 5054/21 Q5 (a), (b), (c)

5(a)(i) C and R marked correctly B1 5(a)(ii) compressions are where pressure is high / particles close together / density is high B1 5(b)(i) 0.75 (Hz) B1 5(b)(ii) speed: no change B1 wavelength: decreases B1 5(b)(iii) at least 3 wavefronts in the shallow region parallel to each other and on correct side of normal M1 refracted towards the surface on correct side of surface and join wavefronts shown A1 5(c) spreading out / bending at an edge / obstacle or spreading out after passing through a (small) gap / hole B1 5054/21 – Mark Scheme PUBLISHED May/June 2025 © Cambridge University Press & Assessment 2025 Page 12 of 15 Question Answer Marks

5054/21	Cambridge O Level – Mark Scheme PUBLISHED	May/June 2025
Question	Answer	Marks
5(a)(i)	C and R marked correctly	B1
5(a)(ii)	compressions are where pressure is high / particles close together / density is high	B1
5(b)(i)	0.75 (Hz)	B1
5(b)(ii)	speed: no change	B1
	wavelength: decreases	B1
5(b)(iii)	at least 3 wavefronts in the shallow region parallel to each other and on correct side of normal	M1
	refracted towards the surface on correct side of surface and join wavefronts shown	A1
5(c)	spreading out / bending at an edge / obstacle or spreading out after passing through a (small) gap / hole	B1

May/June 5054/22

Question 4 · included (a), (b), (c)

(a) Define 'power'.

.....
..... [1]

(b) The efficiency of the motor is 70%.

Calculate the input power to the motor.

input power =W [2]

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(c) The fork-lift truck lifts a load with mass of 50 kg through a vertical distance of 2.3 m.

(i) Calculate the work done in lifting the load.

work done = J [2]

(ii) Calculate the time taken to lift the load.

time taken = s [2]

(a) The force exerted by the flat tyre on the road is 1350 N.

The area of the flat tyre in contact with the road is 0.0070 m^2 .

(i) Calculate the pressure exerted by the flat tyre on the road.

pressure = Pa [2]

(ii) Air is pumped into the tyre to inflate it. This increases the pressure inside. The force exerted by the tyre on the road does not change.

State how the area of the tyre in contact with the road changes.

..... [1]

(b) Inside the inflated tyre, a particle of air collides against the wall of the tyre.

(i) Define 'momentum'.

..... [1]

(ii) Using ideas about momentum, explain why there is a force on the wall of the tyre as the particle collides with it.

.....

 [2]

(c) As the car moves, the temperature of the air inside the tyre increases. The increase in temperature causes an increase in the pressure of the air.

Using ideas about particles, explain why the pressure increases.

.....

 [2]

[Total: 8]

[Turn over]



- 4 A sound wave from a vibrating tuning fork is displayed on the oscilloscope shown in Fig. 4.1. The sound wave is detected using a microphone.

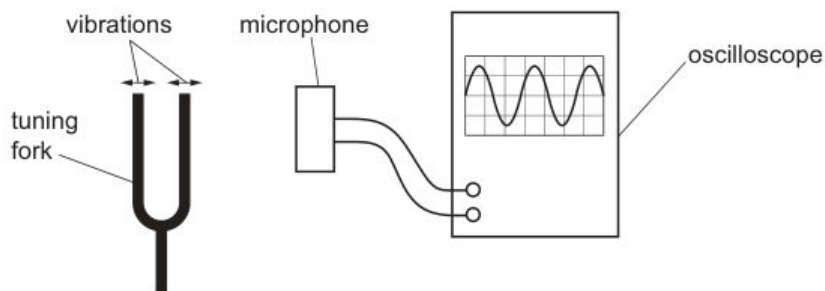


Fig. 4.1

- (a) Describe the motion of the air particles as the sound wave passes from the tuning fork to the microphone.

.....

 [2]

- (b) Fig. 4.2 shows the trace seen on the oscilloscope when the tuning fork vibrates.

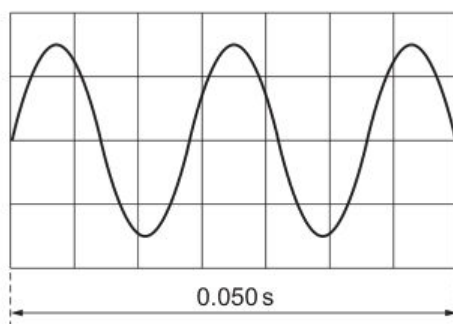


Fig. 4.2

The frequency of the sound is constant.

- (i) State what is meant by 'frequency'.

.....
 [1]

- (ii) The trace shown on Fig. 4.2 is produced in 0.050 s.

Calculate the frequency of the sound wave.

frequency = Hz [2]



- (c) Fig. 4.4 shows three adjacent wavefronts of the sound wave as the wavefronts move towards a gap in a wall created by an open door.

Another student, standing at P on the other side of the wall, hears the sound.

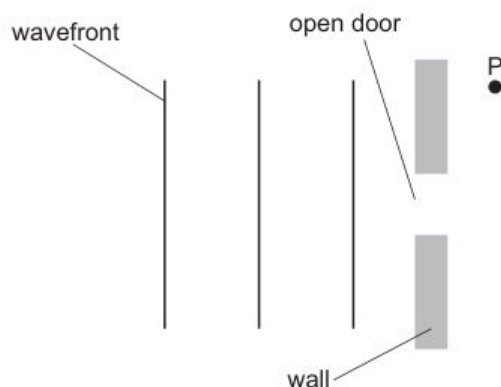


Fig. 4.4

- (i) On Fig. 4.4, draw the three wavefronts after they pass through the gap. [2]
- (ii) State the name of the process involved as the wavefronts pass through the gap and reach P.

..... [1]

[Total: 10]



Mark scheme — 5054/22 Q4 (a), (b), (c)

4(a) motion of particles: backwards and forwards / to and fro / closer and further apart B1 (oscillation) in the direction of travel of the wave / to the right / in direction of microphone / longitudinal / horizontal / parallel to vibrations of tuning fork or mention of compressions and rarefactions B1
 4(b)(i) number of oscillations per second or number of wave(length)s (passing a point) per second B1 4(b)(ii) 2.5 (waves) seen or (1 wave takes) 0.02 (s) or $f = 1/T$ or (number of) waves / time C1 50 (Hz) A1 4(b)(iii) quieter / less volume B1 4(b)(iv) half frequency shown and at least one complete oscillation seen B1 4(c)(i) (at least) three correctly curved wavefronts centered on gap B1 same wavelength as on left and some curving in correct direction B1 4(c)(ii) diffraction B1 5054/22 – Mark Scheme PUBLISHED May/June 2025 © Cambridge University Press & Assessment
 2025 Page 11 of 15 Question Answer Marks

Question	Answer	Marks
4(a)	<i>motion of particles:</i> backwards and forwards / to and fro / closer and further apart	B1
	(oscillation) in the direction of travel of the wave / to the right / in direction of microphone / longitudinal / horizontal / parallel to vibrations of tuning fork or mention of compressions and rarefactions	B1
4(b)(i)	number of oscillations per second or number of wave(length)s (passing a point) per second	B1
4(b)(ii)	2.5 (waves) seen or (1 wave takes) 0.02 (s) or ($f =$) $1 / T$ or (number of) waves / time	C1
	50 (Hz)	A1
4(b)(iii)	quieter / less volume	B1
4(b)(iv)	half frequency shown and at least one complete oscillation seen	B1
4(c)(i)	(at least) three correctly curved wavefronts centered on gap	B1
	same wavelength as on left and some curving in correct direction	B1
4(c)(ii)	diffraction	B1